

A System Perspective on the AI Evaluation Ecosystem

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Abstract

AI evaluation is typically studied as a technical problem—designing better benchmarks, metrics, or test suites. Yet evaluation outcomes do not exist in isolation: they influence model development priorities, market dynamics, capital allocation, and regulatory responses through feedback loops among model developers, evaluators, consumers, policymakers, and funders. These system-level interactions are poorly understood. We make three contributions. First, we present a systems-level framework that maps the AI evaluation ecosystem as a network of actors with strategic incentives and information asymmetries. Second, we operationalize this framework as an LLM-powered agent-based simulation in which each actor reasons over its own objectives and observable state. By systematically ablating ecosystem components and varying regulatory regimes, we generate testable hypotheses about when and how score inflation, validity decay, and market concentration emerge from actor interactions. Third, we develop a diagnostic toolkit for validating LLM-in-the-loop simulations, combining terminating-simulation statistical methods with novel behavioral coherence checks that detect strategy drift, role adherence violations, and prompt sensitivity—addressing a methodological gap as LLM-driven simulations become more common.

1 Introduction

“All models are wrong, but some are useful.”

—George E. P. Box, 1976

Current approaches to evaluating AI systems rely heavily on static benchmarks, leaderboards, and performance metrics. These approaches have helped measure incremental progress on well-defined tasks. However, they provide an incomplete basis for understanding how evaluation practices shape the broader AI ecosystem. Models are not simply measured. Evaluation outcomes influence their behavior, deployment, public perception, and commercial value. Evaluation functions as a strategic intervention that affects how models are trained, marketed, regulated, and trusted. Treating evaluation solely as a technical procedure obscures these system-level dynamics.

A systems perspective emphasizes that evaluation exists within an interconnected ecosystem involving multiple actors. These actors include model developers, auditors, deployers, regulators, end users, and affected communities. They also include funding bodies and

institutions that determine which evaluation methods are incentivized or ignored. Each group engages with evaluation for different reasons. These interactions create feedback loops in which evaluation results influence development priorities, which, in turn, change model capabilities, which then reshape evaluation methods Bommasani et al. [2021], Liang et al. [2023]. For example, safety evaluations can influence what behaviors companies red-team for release Ganguli et al. [2022], Perez et al. [2022], while competitive benchmarks can encourage selective reporting or overfitting to public test sets Thomas and Uminsky [2020], Sainz et al. [2023]. Incident reporting highlights risks or misuses that may have been missed at the pre-deployment stage, and post-market monitoring creates additional feedback loops that shape both evaluation design and model development. These effects are not incidental. They are structural properties of the evaluation ecosystem.

System-level interactions of this kind are already visible in other domains. In healthcare, clinical trials shape which drugs receive investment and regulatory approval Emanuel et al. [2000], Friedman et al. [2010], and such investment incentives can affect the behavior of developers Bates et al. [2022]. Similarly, hospital report cards on metrics such as mortality rates and patient satisfaction influence the treatment behavior of providers Dranove et al. [2003]. In finance, stress testing and audit standards determine capital flows and systemic risk. On content recommendation platforms, evaluation of engagement metrics informs user behavior and platform design. These examples illustrate that evaluation can meaningfully alter the behavior of the systems it measures. AI evaluation likely exhibits similar properties, but we currently lack frameworks to analyze these interactions in a principled way.

Despite their apparent importance, many of these dynamics remain poorly understood. For instance, the degree to which benchmark results affect safety investments, the extent to which closed evaluations alter market competition, and the relationship between evaluation transparency and public trust are open empirical questions. We lack a map of the evaluation ecosystem that identifies key interactions, their strengths, and their consequences. Without this understanding, policy proposals, technical benchmarks, and institutional designs risk addressing symptoms rather than causes. A systems approach is needed to characterize the ecosystem as a whole and to identify where governance mechanisms or new evaluation infrastructure would be most effective.

We aim to develop such a systems perspective. We describe the current evaluation ecosystem, diagnose systemic failure modes arising from misaligned incentives, and identify open questions that require empirical and theoretical study. Our goal is to establish evaluation as a domain that requires systems-level scientific inquiry, not merely incremental metric design. Specifically, this paper makes three contributions:

- **A systems-level framework for the AI evaluation ecosystem.** We map the evaluation ecosystem as a network of actors—model developers, evaluators, consumers, policymakers, funders—with feedback loops in which evaluation outcomes shape development priorities, market dynamics, and regulatory responses (Section 3). This framework identifies structural failure modes that arise from strategic interactions rather than technical limitations of individual benchmarks.
- **An LLM-powered agent-based simulation.** We operationalize the framework as a multi-agent simulation in which each actor (providers, evaluator, consumers,

policymaker, funders, media) is driven by a large language model that reasons over its own objectives and observable state (Section 4). By systematically ablating ecosystem components and varying regulatory regimes, we generate testable hypotheses about when and how score inflation, validity decay, and market concentration arise (Section 6).

- **A diagnostic toolkit for LLM-in-the-loop simulations.** LLM-powered actors introduce stochasticity and potential inconsistency absent from classical agent-based models. We develop a diagnostic protocol that combines terminating-simulation statistical methods (independent replications, common random numbers) with novel behavioral coherence checks (strategy drift, role adherence, prompt sensitivity) and pattern-oriented validation (Section 5). This toolkit addresses a methodological gap: as LLM-driven simulations become more common, the field lacks established practices for validating that emergent dynamics reflect simulation structure rather than LLM artifacts.

A systems-level perspective on AI evaluation should not only describe the ecosystem as it exists today. It should also enable new forms of reasoning about how the ecosystem behaves under pressure, how it will evolve as models and institutions change, and where interventions can be most effective. By focusing on actors, incentives, and information flows, this perspective allows us to anticipate failures that arise only when many parts of the system interact. Benchmark gaming Thomas and Uminsky [2020], Manheim and Garrabrant [2018], contamination Sainz et al. [2023], Balloccu et al. [2024], Magar and Schwartz [2022], or selective disclosure are well-known today, but future failure modes may arise from new strategic dynamics, institutional dependencies, or unexpected market forces. A systems view makes it possible to forecast such failures before they materialize, rather than responding only after problems become visible.

This perspective also helps clarify the role of each actor. Different stakeholders (e.g., model developers, evaluation providers, policymakers, funders, organizational adopters, and individuals) experience different costs, incentives, and constraints. Understanding how these actors influence one another allows us to identify where interventions are likely to be effective. Some interventions require changes in evaluation design. Others require institutional reforms or shifts in funding structures. Without a systems model, it is difficult to determine whether to intervene at the level of benchmarks, access policies, oversight institutions, or market incentives. A detailed ecosystem representation provides a map of leverage points.

Although complex systems research offers general tools for studying interacting components Meadows [2008], AI evaluation requires new paradigms that reflect the specific structure of machine learning workflows and the details of model training. In this domain, optimization is tightly coupled to evaluation signals. New models are developed by targeting benchmarks, and benchmarks evolve in response to the models they measure. This creates forms of performativity MacKenzie [2006], overfitting, and strategic behavior that differ from those in biological, ecological, or economic systems. The distinctive properties of machine learning models, such as high-dimensional training, rapid iteration cycles, and opaque internal mechanisms, produce feedback loops that require specialized modeling approaches. Agent-based simulation offers one such approach Ie et al. [2019]: by instantiating each actor with its own objectives and information, we can observe which systemic outcomes emerge

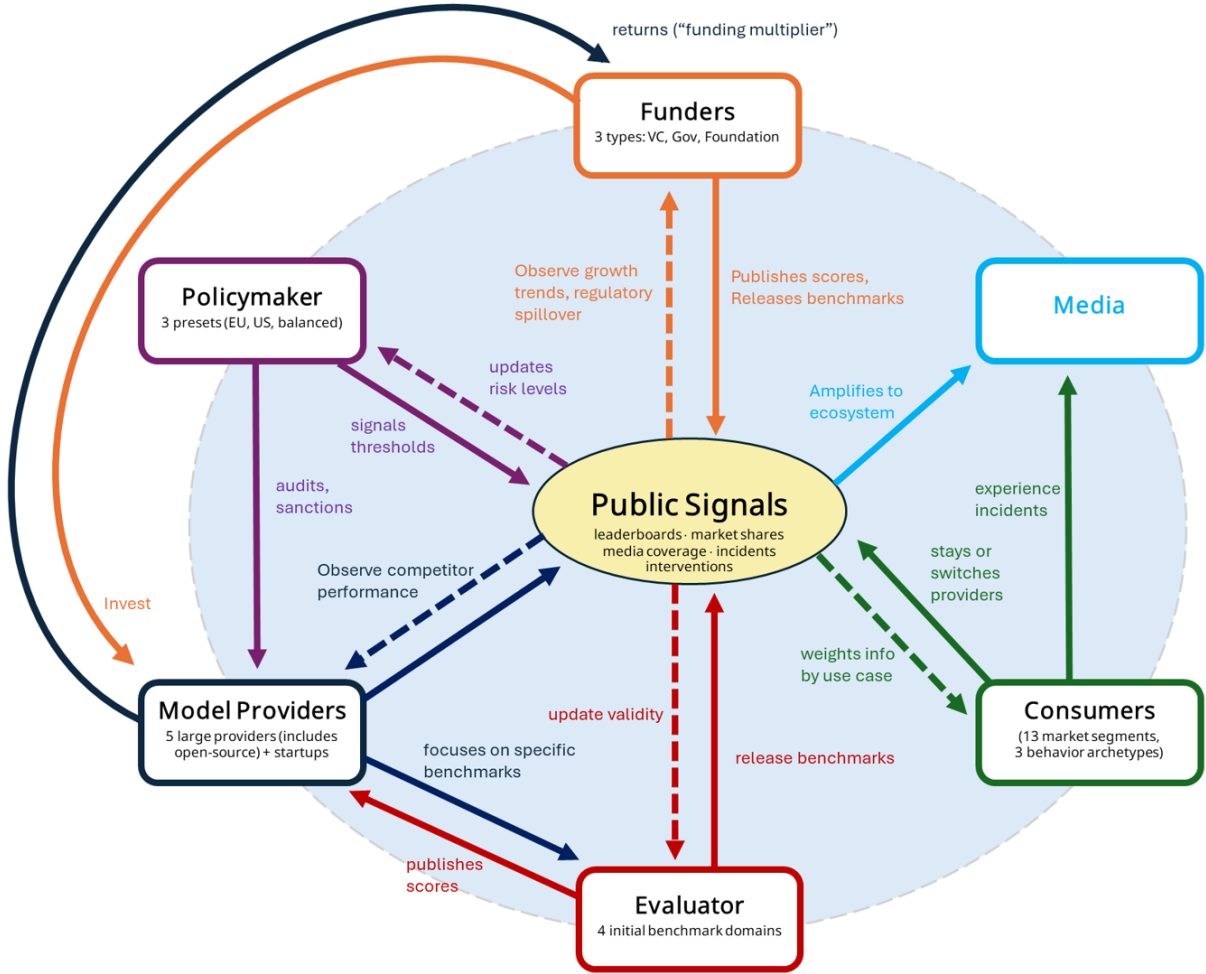


Figure 1: The AI Evaluation Ecosystem showing feedback loops between key actors and stakeholders.

from their interactions and which disappear when specific actors or feedback channels are removed.

2 Related Work

Our work builds on and extends three bodies of literature: AI evaluation and benchmarking, governance and economics of information systems, and systems-level modeling of sociotechnical feedback loops.

2.1 AI Evaluation and Benchmarking

Comprehensive evaluation frameworks such as HELM Liang et al. [2023] and the Foundation Model report Bommasani et al. [2021] have established standards for measuring model

capabilities across diverse tasks. These efforts focus on improving measurement validity, coverage, and reproducibility. Human preference evaluations such as Chatbot Arena Zheng et al. [2023], Chiang et al. [2024] provide dynamic rankings that reflect real-world use cases. While these frameworks advance evaluation methodology, they primarily address *how* to measure models, not *how measurement shapes the ecosystem*. A growing literature examines pathologies in evaluation systems. Work on benchmark gaming Thomas and Uminsky [2020], Manheim and Garrabrant [2018] and contamination Sainz et al. [2023], Balloccu et al. [2024], Magar and Schwartz [2022] identifies technical mechanisms through which optimization pressure degrades benchmark validity. Research on measurement and fairness Jacobs and Wallach [2021], Raji et al. [2021] critiques the construct validity and social implications of evaluation metrics. These contributions document specific failure modes but do not analyze the structural incentives that produce them or the feedback loops through which they propagate across the ecosystem. Our work extends this literature by modeling evaluation as an interconnected system in which technical validity is necessary but insufficient. We examine how competitive dynamics, capital allocation, and regulatory responses interact to produce failures even when individual benchmarks are well-designed.

2.2 Governance and Economics of Information Systems

Our public goods framing draws on economic analysis of information security Anderson and Moore [2006], Moore et al. [2010], which demonstrates that market incentives systematically underprovide cybersecurity due to externalities, free-riding, and misaligned liability. We extend this framework to AI evaluation, arguing that trustworthy evaluation exhibits similar public goods characteristics: non-excludability (evaluation results benefit all actors), underinvestment (no single actor captures the full value of rigorous testing), and systemic risk from concentration. Work on AI governance Papernot et al. [2022], Whittlestone et al. [2019], Veale and Borgesius [2021] and trustworthy AI development Brundage et al. [2020] proposes mechanisms for accountability and transparency, including model cards Mitchell et al. [2019], datasheets Gebru et al. [2021], and third-party auditing Raji et al. [2020]. These interventions address individual model releases but do not examine ecosystem-level dynamics such as competitive benchmark optimization, selective disclosure, or market concentration. Recent regulatory frameworks such as the EU AI Act European Parliament and Council [2024] and NIST AI Risk Management Framework National Institute of Standards and Technology [2023] mandate evaluation requirements but do not yet address the feedback loops that undermine evaluation integrity when it becomes a competitive signal. We contribute by analyzing evaluation through the lens of market failures and institutional design, drawing parallels with clinical trials Emanuel et al. [2000], Friedman et al. [2010] and aviation safety systems Hollnagel [2014], Leveson [2004] to identify governance mechanisms that support both innovation and safety.

2.3 Systems-Level Modeling and Performativity

Our approach is informed by systems thinking Meadows [2008] and work on performativity in algorithmic systems MacKenzie [2006], which examines how measurement and optimization co-evolve. These perspectives emphasize feedback loops, emergent properties, and strategic

responses to measurement. However, prior work has not applied these frameworks systematically to AI evaluation or developed computational models of ecosystem dynamics. Our agent-based simulation extends methodologies from generative agent research by modeling strategic interactions among heterogeneous actors (providers, evaluators, consumers, policymakers, funders) with asymmetric information and competing objectives. Unlike prior simulations of algorithmic systems, ours explicitly models the gap between benchmark scores and true capability, allowing us to study score inflation, validity decay, and market concentration as emergent phenomena. We also build on participatory and pluralistic approaches to evaluation Birhane et al. [2022], Sloane et al. [2022], which emphasize that no single benchmark captures diverse stakeholder needs. Our simulation demonstrates that pluralism alone is insufficient: even with multiple benchmarks and mission-driven funding, strategic optimization and market dynamics produce systematic failures.

A central design question in generative agent-based modeling is how much behavioral flexibility to delegate to LLM actors versus encoding in closed-form rules. Critical reviews argue that LLM agents exacerbate rather than resolve classical ABM challenges: their black-box nature obscures causal mechanisms, and their tendency toward distribution collapse produces an “average persona” that underrepresents behavioral variance Larooij and Törnberg [2025]. Wu et al. [2025] formalize this concern, showing that LLM-generated behaviors satisfy mean alignment with human populations but fail to reproduce the variance required for heterogeneous social dynamics, and propose that claims should be constrained to collective-level qualitative patterns when variance is insufficient. Our simulation navigates this tension by restricting LLM actors to strategic planning decisions (investment allocation, pricing) while grounding market mechanics, scoring, and state transitions in deterministic rules. This hybrid design preserves the expressiveness of natural-language reasoning—where actors must weigh multidimensional tradeoffs that resist compact closed-form specification—while keeping the simulation’s causal structure transparent and reproducible. We further mitigate distribution-collapse risk through behavioral coherence checks (Section 5) and pattern-level rather than point-estimate claims (Section 6). Complementary evidence that LLMs are well-suited to play strategic actors comes from Kim et al. Kim et al. [2026], who show that reasoning models internally generate “societies of thought”—multi-agent debates among implicit personas—suggesting that the deliberative structure our simulation externalizes across actors may mirror the internal computation LLMs already perform.

A growing body of work uses LLM-powered agents as participants in social simulations, but validation methodology remains nascent. Park et al. Park et al. [2023] evaluate generative agents via human believability interviews and component ablations, without multi-run statistical analysis. Aher et al. Aher et al. [2023] validate by replicating classic human-subject experiments (Ultimatum Game, Milgram) and comparing aggregate statistics against published human data, uncovering systematic “hyper-accuracy” distortions. Gao et al. Gao et al. [2023] validate a social-network simulator at both micro (individual next-state prediction) and macro (aggregate pattern matching) levels against real social-media traces. Ghaffarzadegan et al. Ghaffarzadegan et al. [2024] introduce prompt-sensitivity sweeps as a robustness check for generative agent-based models, and Manning et al. Manning et al. [2024] ground LLM simulations in structural causal models that enable formal hypothesis testing. Most recently, Bao et al. Bao et al. [2025] propose a two-stage protocol—first replicating established experimental findings, then generating novel predictions—that represents the

most systematic validation effort to date. Despite this progress, critical reviews identify persistent challenges: LLM agents exhibit distribution collapse (approximating population means while underrepresenting variance), temporal instability across model API updates, and low behavioral diversity relative to human participants Chen et al. [2025], Bassignana et al. [2025]. Our diagnostic protocol (Section 5) addresses these gaps by grounding LLM simulation analysis in the terminating-simulation inference framework of Law Law [2015] and the ABM verification-and-validation standards of Sargent Sargent [2013], complemented by LLM-specific coherence checks that the existing literature largely omits.

3 The AI Evaluation Ecosystem

Evaluation does not occur in isolation. It arises from interactions among diverse actors who have different goals, access to information, resource constraints, and decision-making horizons. Each actor engages with evaluation for strategic reasons. Their observable behavior creates feedback loops that shape both model development and the evolution of evaluation methods.

Critically, the evaluation signals that drive these interactions extend well beyond public benchmarks. Quantitative evaluations—standardized benchmarks Liang et al. [2023], leaderboard scores Zheng et al. [2023], accuracy metrics, and red-team challenge numbers Ganguli et al. [2022]—feed directly into optimization loops. But qualitative signals—expert analyses, incident reports, interpretability studies, and narrative failure summaries—also shape the ecosystem by guiding public perception, policymaker attention, and organizational risk assessments Raji et al. [2021]. Evaluation signals further differ along other dimensions: public vs. private (leaderboards vs. internal audits), personalized vs. universal (domain-specific workflows vs. broad capability benchmarks Birhane et al. [2022], Sloane et al. [2022]), and formal vs. emergent (designed test suites vs. spontaneous user feedback and real-world failure reports). The ecosystem behaves very differently depending on which of these signals circulate, which are visible, and which can be optimized against. A narrow focus on public benchmarks alone would miss much of the feedback that shapes model development and policy responses.

Below, we outline several central players in this ecosystem, emphasizing what information they access, how they form beliefs, and what actions they take in response. We then describe the feedback loops that connect them.

3.1 Actors

Individual Consumers Individual consumers interact directly with model outputs and publicly available evaluation results. They typically discover models through leaderboard rankings, word-of-mouth popularity, or curated interfaces such as marketplaces and open-source repositories. Their state includes personal needs, budget, prior beliefs about model capability, and trust in specific evaluation sources. Individuals choose among models, query them, participate in public comparisons such as LM Arena, or decide not to adopt models at all. These choices contribute to model reputation and usage statistics, which feed back into market demand and future evaluation priorities.

Organizational Consumers Organizational consumers such as hospitals, firms, platforms, and government agencies face different constraints. Decisions involve integration costs, regulatory exposure, stability requirements, and responsibility for downstream impacts on clients or patients. These organizations observe leaderboard rankings, compliance certifications, model performance reports, and costs of deployment at scale. Their internal state includes budget constraints, institutional goals, available expertise, and integration friction. Organizations may build internal benchmarking pipelines, combine multiple models into compound systems, commission evaluations, or develop lightweight alternative models. Their deployment decisions affect demand signals, institutional norms, and sources of proprietary evaluation data.

Policymakers and Regulators Policymakers observe evaluation results through public leaderboards, technical reports, confidential disclosures from model providers, and third-party audits. Their internal state includes policy priorities, risk assessments, and the capacity to interpret technical evaluations Whittlestone et al. [2019], Veale and Borgesius [2021]. Policymakers can require specific evaluations for compliance, commission new evaluation programs, negotiate with developers, and enforce sanctions or disclosure requirements. They also shape norms by signaling regulatory direction or defining thresholds for intervention. Their decisions influence what evaluations become mandatory, which methods become standardized, and whether evaluation capacity is built inside governments or outsourced to third parties.

Model Providers Model providers such as OpenAI, Google, Anthropic, and open-source consortia observe both internal and external information. They have private access to fine-grained model behavior, proprietary user data, internal red-team results, and commercial metrics. They also observe competitor models, public benchmarks, popularity signals, and policy requirements. Their internal state includes financial resources, compute, data access, cost structures, investor priorities, and internal metrics. Providers act by releasing models, adjusting training strategies to improve benchmark performance, lobbying policymakers, withdrawing models from deployment, or investing in research directions shaped by evaluation expectations. These decisions determine which capabilities progress and which remain under-resourced.

Evaluation Providers Evaluation providers include academic groups, independent labs, industry consortia, and emerging third-party testing institutions such as HELM Liang et al. [2023], METR Shevlane et al. [2023], and domain-specific auditors Raji et al. [2020]. They observe demand from developers and policymakers, funding availability, benchmark popularity, and emerging risks. Their internal state includes resource constraints, risk priorities, and conceptual models of what should be evaluated. They generate new test sets, define evaluation metrics, publish benchmark scores, and retire outdated evaluations. Since evaluations can steer development priorities, evaluation providers indirectly influence which directions receive investment. They may also monetize evaluation tools, infrastructure, or proprietary datasets.

Funders and Capital Allocators Funders include philanthropic organizations, venture capital firms, government agencies, research foundations, and mission-driven grantmakers.

They observe performance signals from leaderboards, institutional reputation, scientific impact, and market adoption. Their internal state includes return expectations, mission alignment, expert committees, and perceived importance of different research areas. Funders act by awarding grants, shaping calls for proposals, defining priority areas, and selecting institutions to support. Their decisions drive which evaluations are built, what research pipelines are financially viable, and what capabilities gain early investment. Funding decisions create strong directional pressure on both evaluation development and model development.

3.2 Feedback Loops of Evaluation

The AI evaluation ecosystem contains multiple feedback loops in which evaluation outcomes drive strategic responses by developers, institutions, and users. These responses change the underlying system, which then alters what future evaluations measure. Understanding evaluation as part of a feedback system clarifies why models improve rapidly on benchmarks while remaining unreliable in real-world deployment, and why evaluation practices tend to converge toward narrow axes of progress.

One of the most direct feedback loops occurs between public benchmarks and model training pipelines. When benchmarks become signals of capability, model providers invest effort in improving performance on those benchmarks. This includes data collection targeted at benchmark domains, specialized fine-tuning, and repeated prompting experiments to discover stable strategies. Success on a benchmark then drives adoption, press attention, and investor interest, which further incentivizes benchmark-oriented optimization. This loop can accelerate progress but also narrows development toward what is easy to measure.

A second feedback loop involves market share and competitive pressures. Leaderboards influence adoption choices among individuals and organizations, especially when the cost of switching between models is low. Providers respond by prioritizing evaluation categories that drive market differentiation, such as math performance, reasoning scores, or agentic benchmarks, while deprioritizing dimensions—such as safety—that do not directly drive adoption or investment. This dynamic has been formalized as a “race to the precipice” in which competitive pressure systematically discourages safety investment Armstrong et al. [2016], and recent empirical work suggests that many AI safety benchmarks are highly correlated with general capabilities, enabling providers to claim safety progress through capability improvements alone Ren et al. [2024]. Public performance signals can also shape investor perceptions, which affect capital allocation and staffing. In these cases, evaluations are not neutral diagnostics. They become strategic instruments that coordinate expectations in the marketplace.

A third feedback loop emerges around secrecy and selective disclosure. Providers may internalize that high performance on public benchmarks drives market credibility, while poor results may trigger negative press or regulatory scrutiny. As a result, providers may avoid participating in evaluations where outcomes could be unfavorable. They may also withhold model access or restrict evaluation terms to prevent external benchmarking. This incentives structure can lead to a pattern where high-performing results are public and low-performing results remain private. It also shifts evaluation power toward internal testing regimes that are difficult to audit.

Feedback loops can also drive technical distortions. Over-reliance on static test sets creates

incentives for contamination, where training data inadvertently or deliberately overlaps with benchmark content Sainz et al. [2023], Balloccu et al. [2024], Magar and Schwartz [2022]. Even without exact memorization, models can learn benchmark-specific artifacts or heuristics. Fine-tuning pipelines can optimize for benchmark quirks rather than construct-valid performance. These effects resemble overfitting in machine learning but occur at the system level, driven by institutional incentives rather than optimization alone Thomas and Uminsky [2020], Manheim and Garrabrant [2018]. Contamination reduces the diagnostic value of benchmarks and increases the cost of designing evaluations that measure general capability.

Empirical evidence of these dynamics is already emerging. Developers have announced improvements targeted at specific benchmarks shortly after those benchmarks gained visibility. Models show abrupt performance gains on public datasets while remaining inconsistent on semantically equivalent but unseen variants Sainz et al. [2023]. New evaluation formats such as red-teaming challenges and agentic stress tests have influenced internal investment in safety and interpretability research Ganguli et al. [2022], Shevlane et al. [2023]. In competitive settings like Chatbot Arena Zheng et al. [2023], Chiang et al. [2024], performance swings after fine-tuning cycles reflect rapid adaptation to evaluation conditions rather than broad capability growth.

Across these examples, evaluation functions as part of a coupled system. Benchmarks shape models. Models shape evaluation design. Both shape market behavior and policy responses. Recognizing these feedback loops is essential for designing evaluation methods that remain informative even when they become targets of optimization.

4 Evaluation Ecosystem Simulator

To study how emergent failures arise from strategic interactions in the AI evaluation ecosystem, we developed a multi-agent simulation modeling six provider types, three funder types, a policymaker, a media actor, and 39 consumer segments. The simulation operationalizes the feedback loops described in Section 3 and provides a controlled environment for reasoning about which structural conditions enable or suppress specific failure modes. Following the design philosophy of configurable simulation platforms such as RecSim Ie et al. [2019], we use the simulation as a hypothesis generator: the results surface candidate dynamics and suggest pathways worth investigating empirically, rather than proving that specific failures will occur in real systems.

4.1 Simulation Design

The simulation models six AI model providers competing across eight benchmark domains: coding, reasoning, math, safety, writing, medical, legal, and finance. Provider identities are anonymized to prevent LLM reasoning from being biased by real-world company reputations; the six providers are Orion Labs, Apex AI, Genesis Systems, Mirage AI, Spark AI, and OpenCore (an open-source provider). Each provider is characterized by:

- **True capability:** An unobserved ground-truth measure of model quality, held exclusively by the simulation and structurally inaccessible to any actor or LLM prompt.

- **Investment portfolio:** Allocation across four categories—fundamental research ($1.5\times$ capability efficiency; breakthrough-eligible), training optimization ($1.0\times$ efficiency), evaluation engineering (directly inflates scores via benchmark exploitability without improving true capability), and safety alignment (reduces incident probability; no direct capability effect).
- **Strategic profile:** Distinct organizational priorities. Apex AI prioritizes safety and fundamental research (40% safety, 30% research), while Spark AI concentrates on evaluation engineering (45%), and OpenCore operates as an open-source provider that accelerates benchmark contamination via a $1.8\times$ contamination multiplier while remaining exempt from regulatory sanctions and VC funding.
- **Benchmark focus (benchmark_focus):** Each provider maintains a weight vector that routes its evaluation engineering budget across benchmark domains, allowing specialists to concentrate gaming effort on targeted benchmarks.

Benchmarks are modeled with two independent degradation parameters: **validity** (α), measuring the correlation between published scores and true capability (signal quality), and **exploitability** (β), measuring how much evaluation engineering investment can systematically inflate scores above true capability (the gaming lever). The scoring model is $\hat{s} \sim \mathcal{N}(c + \beta e, \sigma^2/\alpha)$, where c is true capability and e is evaluation engineering. Both parameters degrade under Goodhart pressure: each round, α decreases and β increases proportional to the ecosystem-wide mean evaluation engineering. Benchmarks retire when any provider reaches a score of 0.90 and are replaced from a predefined domain sequence.

The consumer market comprises 39 segments defined by 13 use-case profiles crossed with six behavioral archetypes: *leaderboard_follower*, *experience_driven*, and *cautious* (individual consumers); and *enterprise_cautious*, *enterprise_growth*, and *enterprise_established* (organizational consumers). Archetypes vary in leaderboard trust, switching friction, and cost sensitivity. Three funder types (VC, government, foundation) allocate capital based on observed scores, market traction, and perceived gaming risk, producing a **funding_multiplier** $\in [1.0, 2.0]$ on provider R&D efficiency; VC funders skip open-source providers entirely (no equity model). A policymaker actor monitors score inflation and consumer dissatisfaction, applying a graduated escalation ladder (investigation $\rightarrow$ public warning $\rightarrow$ benchmark mandate $\rightarrow$ compliance audit $\rightarrow$ sanctions). A media actor publishes coverage after each scoring round, modulating consumer belief-update rates, policymaker risk signals, and funder gaming suspicion.

New startup entrants can spawn mid-simulation with effective entry probability $p_{\text{eff}} = p_{\text{base}} \times (1 - \text{BTE}_{\text{composite}})$, where $\text{BTE}_{\text{composite}}$ is a per-round barrier-to-entry index comprising market concentration, capability gap, funding lock-in, and consumer lock-in. Entrants start with capability equal to 75% of the best open-source provider, safety alignment of 0.10, and a lean cost advantage of 0.38.

Information access is enforced via a three-tier visibility system: public state (published scores, leaderboard, regulatory interventions) is accessible to all actors; private state (beliefs, strategies, satisfaction history) is accessible only to the agent itself; and ground truth (true capability, true satisfaction) is held exclusively by the simulation and is structurally inaccessible to any actor or LLM prompt.

The policymaker is calibrated to three regulatory presets: EU precautionary (intervention threshold 0.35; fine multiplier 0.35; sanction duration 4 rounds), balanced, and US light-touch (intervention threshold 0.75; fine multiplier 0.10; sanction duration 2 rounds). At 60% market share, the EU fine formula produces a 16.8% R&D efficiency reduction per round versus 1.8% under the US preset. Open-source providers are exempt from both sanction conditions and market concentration reviews.

Figure 2 illustrates the round-order interaction structure and key feedback loops. Algorithm 1 specifies the full round protocol.

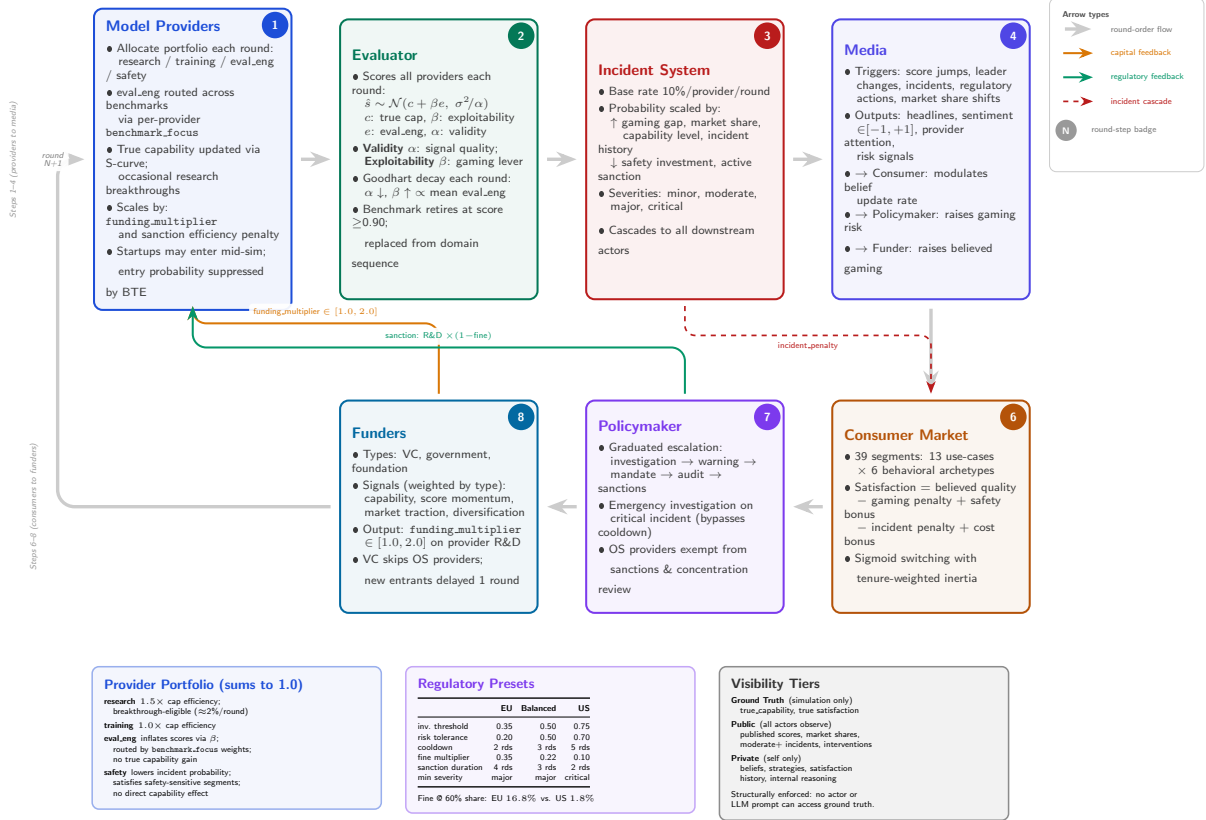


Figure 2: Round-order interaction structure of the simulation. Providers update capabilities and scores first; incidents, media, consumers, policymakers, and funders then observe and respond in sequence. Capital and regulatory feedback loops close at the start of the next round.

4.2 Design Rationale

Key design choices were made to reflect the structure of real evaluation ecosystems and to enable the simulation to produce differentiated, interpretable dynamics. We summarize the main decisions here; full parameter derivations are in the Appendix.

Algorithm 1 Simulation round protocol. Each round proceeds through eight sequential phases. Actors observe only public and own-private state; ground truth ($\tilde{c}$) is held by the simulation and never exposed to any actor prompt.

Require: Providers $\mathcal{P}$, Evaluator $\mathcal{E}$, Benchmarks $\mathcal{B}$, ConsumerMarket $\mathcal{M}$, Policymaker π , Funders $\mathcal{F}$, Media μ , rounds T , ground truth $\tilde{c}$

- 1: **Initialise** provider capabilities $\tilde{c}_p$, strategies $\mathbf{x}_p$, benchmark parameters $(\alpha_b, \beta_b, \sigma_b)$, consumer segments, funder budgets, regulatory preset
- 2: **for** $t = 1, \dots, T$ **do**
 - % Phase 1: Market entry*
 - 3: With probability $p_{\text{entry}}(1 - \text{BTE}_t)$, spawn startup $\rightarrow \mathcal{P} \leftarrow \mathcal{P} \cup \{p_{\text{new}}\}$
 - % Phase 2: Provider planning & capability update*
 - 4: **for** each provider $p \in \mathcal{P}$ **do**
 - 5: $\mathbf{o}_p \leftarrow \text{GETPUBLICCONTEXT}(p, \mathcal{H}_{t-1})$ $\triangleright$ scores, market share, sanctions, incidents
 - 6: $\mathbf{x}_p \leftarrow \text{PLAN}(\mathbf{o}_p, \mathbf{s}_p^{\text{priv}})$ $\triangleright$ LLM or heuristic $\rightarrow$ portfolio allocation
 - 7: $f_p \leftarrow$ funding multiplier from $\mathcal{F}$ at $t-1$; $\eta_p \leftarrow$ sanction penalty from π at $t-1$
 - 8: $\tilde{c}_p \leftarrow \tilde{c}_p + \Delta(\mathbf{x}_p, f_p, \eta_p)$ $\triangleright$ S-curve capability growth
 - 9: **end for**
 - % Phase 3: Evaluation & benchmark dynamics*
 - 10: $\hat{\mathbf{s}} \leftarrow \mathcal{E}.\text{ScoreAll}(\mathcal{P}, \tilde{c}, \mathcal{B})$ $\triangleright \hat{s}_p \sim \mathcal{N}(\tilde{c}_p + \beta_b e_p, \sigma_b^2 / \alpha_b)$
 - 11: Update α_b, β_b based on aggregate eval engineering $\triangleright$ Goodhart decay
 - 12: Retire saturated benchmarks; introduce new benchmark if cooldown elapsed
 - 13: Publish leaderboard $L_t \leftarrow \text{sort}(\hat{\mathbf{s}})$
 - % Phase 4: Provider observation & reflection*
 - 14: **for** each provider $p \in \mathcal{P}$ **do**
 - 15: $p.\text{Observe}(\hat{s}_p, \hat{s}_{\neg p})$; $p.\text{Reflect}()$ $\triangleright$ update beliefs about own & competitor capability
 - 16: **end for**
 - % Phase 5: Incident generation*
 - 17: $I_t \leftarrow \text{GenerateIncidents}(\mathcal{P}, \tilde{c}, \hat{\mathbf{s}}, \text{market shares})$
 - % Phase 6: Downstream actors observe & act*
 - 18: $m_t \leftarrow \mu.\text{ObserveAndPublish}(L_t, \mathcal{B}, I_t)$ $\triangleright$ media coverage
 - 19: $d_t \leftarrow \mathcal{M}.\text{Update}(L_t, \tilde{c}, m_t, I_t)$ $\triangleright$ satisfaction, market shares
 - 20: $r_t \leftarrow \pi.\text{ObserveReflectPlan}(L_t, d_t, m_t, I_t)$ $\triangleright$ regulatory intervention
 - 21: $\phi_t \leftarrow \mathcal{F}.\text{ObserveReflectPlan}(L_t, d_t, r_t, m_t, I_t)$ $\triangleright$ funding allocations
 - % Phase 7: Log round*
 - 22: $\mathcal{H}_t \leftarrow (\hat{\mathbf{s}}, \tilde{c}, \mathbf{x}, \mathcal{B}, d_t, r_t, \phi_t, I_t, m_t, \text{traces})$
 - 23: **end for**

An important clarification about the role of assumptions in the simulation: the model

Table 1: Summary of simulation design hyperparameters. See the Appendix for full derivations.

Parameter	Value	Parameter	Value
<i>Ecosystem structure</i>		<i>Investment efficiency</i>	
Providers	6	Fundamental research	$1.5\times$
Benchmark domains	8	Training optimization	$1.0\times$
Consumer segments	39	OpenCore contamination	$1.8\times$
Funder types	3		
Regulatory presets	3		
<i>Provider initialization</i>		<i>Benchmark dynamics</i>	
Init. capability (mean)	0.25	Scoring model	$\hat{s} \sim \mathcal{N}(c + \beta e, \sigma^2/\alpha)$
Init. capability (range)	[0.21, 0.27]	Retirement threshold	0.90
Entrant capability	$0.75\times$ best OSS	Degradation horizon	8–12 rounds
Entrant safety alignment	0.10		
Entrant cost advantage	0.38		
<i>Regulatory presets (EU / Bal. / US)</i>		<i>Funding & market entry</i>	
Intervention threshold	0.35 / 0.50 / 0.75	Funding multiplier range	[1.0, 2.0]
Fine multiplier	0.35 / 0.22 / 0.10	Base incident rate	0.10
Sanction duration (rounds)	4 / 3 / 2	Entry probability	$p_{\text{base}} \times (1 - \text{BTE}_c)$

assumes that providers *can* invest in evaluation engineering that inflates scores without proportionally improving true capability—a mechanism grounded in documented practices such as benchmark-specific fine-tuning, selective checkpoint submission, and data contamination Sainz et al. [2023], Balloccu et al. [2024]. The simulation does not assume that providers *will* adopt this strategy, nor does it predetermine the ecosystem-level consequences. Whether gaming becomes a stable equilibrium, which actors amplify or suppress it, and how it interacts with regulation, funding, and consumer behavior are all emergent outcomes that vary across experimental conditions. The ablation design is specifically constructed to test these contingencies: if gaming were an artifact of parameter tuning rather than a structural property, removing actors or changing regulatory regimes would eliminate it. As reported in Section 6, it persists across all eleven conditions, but its magnitude and character vary substantially—findings that are not entailed by the assumption alone.

Benchmark domain selection. The eight domains span the established landscape of public LLM evaluation suites and the sectors where AI deployment is most commercially significant and socially consequential. Coding is anchored by HumanEval Chen et al. [2021] and SWE-bench Jimenez et al. [2024]; reasoning and general knowledge correspond to MMLU Hendrycks et al. [2020]; math reflects GSM8K Cobbe et al. [2021] and MATH Hendrycks et al. [2021]; and safety is informed by TruthfulQA Lin et al. [2021] and WMDP Li et al. [2024]. Medical, legal, and finance represent the three highest-stakes professional verticals in current AI deployment, where benchmark performance directly influences procurement decisions Thirunavukarasu et al. [2023], Katz et al. [2024], Wu et al. [2023]. Critically, these domains differ in gaming susceptibility: coding and math benchmarks are graded by exact execution or symbolic match

and are more susceptible to data contamination Sainz et al. [2023], Balloccu et al. [2024], whereas medical and legal evaluation often requires open-ended generation that is harder to game cheaply. This heterogeneity is a deliberate design feature that produces differentiated gaming fingerprints across providers, consistent with the argument that benchmark diversity is necessary to detect selective optimization Kiela et al. [2021].

Provider profiles and anonymization. Initial true capability values (mean 0.25, range 0.21–0.27) are intentionally low relative to the $[0, 1]$ scoring scale, reflecting that even frontier models leave substantial room for improvement on hard benchmarks Stanford HAI [2024]. This ensures benchmark saturation occurs organically through gaming inflation rather than from unrealistically high starting capability. Provider names are anonymized (Orion Labs, Apex AI, Genesis Systems, Mirage AI, Spark AI, OpenCore) to prevent LLM-powered actor agents from importing real-world reputational priors into in-simulation decisions, which would confound the experimental manipulation of simulated parameters.

Regulatory preset calibration. The EU and US presets reflect the structural differences between the EU AI Act European Parliament and Council [2024] and the US voluntary-commitments approach in the NIST AI RMF National Institute of Standards and Technology [2023] and Executive Order 14110 The White House [2023]. The EU preset’s intervention threshold of 0.35 reflects the Act’s ex-ante, tiered-risk architecture, under which regulators are expected to act before widespread harm materializes (Article 6). The US preset’s threshold of 0.75 reflects the post-hoc, voluntary character of EO 14110. The fine multipliers (EU: 0.35, US: 0.10) are calibrated so that at 60% market share the EU fine reduces R&D efficiency by approximately 16.8% per round versus 1.8% for the US—broadly consistent with the EU AI Act’s ceiling of 3% of global annual turnover for violations (Article 101) versus the consent-decree-only posture of US FTC actions.

Validity decay and evaluation engineering returns. The benchmark degradation model—in which validity α decreases and exploitability β increases each round proportional to ecosystem-wide gaming effort—operationalizes the empirical observation that benchmarks lose diagnostic value as they become optimization targets. This is a direct application of Goodhart’s Law to evaluation dynamics Manheim and Garrabrant [2018], Thomas and Uminsky [2020]. The rate of decay is not calibrated to a specific empirical benchmark lifecycle, as no such longitudinal dataset exists. Instead, the parameters are set so that benchmarks degrade over approximately 8–12 rounds of sustained gaming pressure before reaching retirement thresholds, producing turnover rates qualitatively consistent with the observed lifecycle of public benchmarks (e.g., the diminishing informativeness of MMLU Hendrycks et al. [2020] and the introduction of MMLU-Pro as a replacement). The separation of validity and exploitability is a deliberate modeling choice: a benchmark can have low validity due to noise or poor construct design even without gaming, while a highly exploitable benchmark may still correlate with capability in the absence of strategic effort. This distinction matters because the two failure modes call for different interventions—improving test design versus limiting optimization pressure. Evaluation engineering investment translates to score inflation through the βe term in the scoring model. The key assumption is that providers can invest in strategies

that raise benchmark scores without proportionally improving true capability—consistent with documented practices such as benchmark-specific fine-tuning, prompt engineering for test formats, and selective checkpoint submission Sainz et al. [2023], Balloccu et al. [2024]. Because the simulation serves as a hypothesis generator rather than a predictive model, the specific magnitudes matter less than the qualitative dynamics they produce; the ablation design tests whether conclusions are robust to the presence or absence of structural ecosystem features rather than to specific parameter values.

Incident probability model. The base incident rate of 10% per provider per round is a modeling parameter rather than a direct empirical estimate, as no published database currently provides per-system per-period incident rates with reliable denominators Partnership on AI [2021], OECD [2024]. Six scaling factors implement theoretically motivated relationships: safety investment captures the diminishing-returns structure in cybersecurity investment models Anderson and Moore [2006]; the gaming-gap factor encodes the hypothesis that score inflation implies de-emphasized safety-relevant capability—the core Goodhart mechanism applied to risk Manheim and Garrabrant [2018]; market-share and capability-level factors implement an exposure model; and an incident-history escalation factor reflects that organizations with prior incidents show elevated recurrence before reforms take effect Weidinger et al. [2023].

5 Simulation Diagnostics

Our simulation is an agent-based model (ABM) by architecture—heterogeneous strategic agents interacting through local rules that produce emergent macro-dynamics—and a *terminating simulation* by analysis type Law [2015]: initial conditions are fixed and substantively meaningful, the horizon is finite, and the full trajectory (not a long-run steady-state average) is the object of inference. This classification determines the appropriate diagnostic framework. Unlike steady-state simulations or MCMC samplers, terminating simulations require no ergodicity or stationarity assumptions; the formal basis for inference is simply independence across replications and the classical central limit theorem Law [2015]. We complement this statistical framework with ABM-specific validation practices drawn from Sargent’s verification-and-validation (V&V) protocol Sargent [2013] and the pattern-oriented modeling approach of Grimm et al. Grimm et al. [2005], and add diagnostics specific to LLM-powered actors that have no classical precedent. We outline the diagnostic protocol below; a summary of results appears in Section 6.2 and full tables in Appendix 10.1.

Independent replications and statistical inference. The standard method for analyzing terminating simulations is to execute N independent replications of each experimental condition, yielding N i.i.d. copies of every outcome trajectory Law [2015]. We report inter-run means, standard deviations, and 95% confidence intervals for all primary outcome variables (mean true capability, aggregate gaming intensity, market concentration, consumer welfare). Qualitative conclusions are accepted only when they hold across the majority of replications, not merely in a single representative run. When comparing paired conditions (e.g., EU versus US regulation), we apply common random numbers (CRN) by matching

seeds across conditions. CRN controls the simulation’s internal random number stream (incident sampling, market dynamics, benchmark noise), reducing inter-replication variance in paired-difference tests Law [2015]. However, LLM-powered actors introduce server-side sampling stochasticity that seeds cannot control: even at temperature zero, API responses are not perfectly deterministic. CRN therefore provides partial—but not complete—variance reduction relative to fully seed-controlled simulations. We complement CRN-paired tests with heuristic-mode replications (where all randomness is seed-governed) to establish the CRN variance-reduction baseline.

Trajectory inspection and initialization bias. For each replication, we plot round-by-round traces of key aggregate quantities: mean true capability, mean evaluation-engineering investment, benchmark validity (α), Herfindahl–Hirschman index of market concentration, and consumer satisfaction. In terminating simulations the initial transient is part of the phenomenon of interest rather than a nuisance to be discarded Law [2015]; we therefore report statistics over the full trajectory. Trace plots nonetheless serve to detect pathological behavior such as absorbing states, divergence, or oscillatory instability that would indicate model misspecification rather than meaningful dynamics.

Sensitivity analysis. To assess whether qualitative findings are artifacts of specific parameter choices, we conduct one-at-a-time perturbation sweeps over key hyperparameters: initial true capability ($\pm 20\%$), base incident rate ($\pm 50\%$), evaluation-engineering efficiency (β scaling), benchmark retirement threshold, and funding multiplier bounds. For each perturbation, we re-run the full replication set and compare outcome distributions against the baseline. A finding is considered robust if its direction and qualitative magnitude persist across the perturbation range. This follows the sensitivity-analysis component of Sargent’s V&V framework Sargent [2013].

Pattern-oriented validation. Rather than calibrating the simulation to reproduce specific numerical targets, we adopt the pattern-oriented modeling approach Grimm et al. [2005], Windrum et al. [2007]: the simulation is considered valid if it simultaneously reproduces multiple qualitative patterns observed in real evaluation ecosystems (e.g., benchmark saturation and turnover, score inflation preceding capability plateaus, market concentration under weak regulation). A model that reproduces a single pattern may do so for artifactual reasons; simultaneous reproduction of structurally independent patterns provides stronger evidence that the underlying mechanisms are correctly specified Grimm et al. [2005].

Behavioral coherence of LLM actors. LLM-powered agents introduce a source of stochasticity and potential inconsistency absent from classical ABMs with closed-form decision rules. We audit actor coherence by logging each agent’s decision rationale alongside its actions and checking for: (i) strategy drift, where an agent’s revealed investment allocation diverges from its stated reasoning over successive rounds; (ii) prompt sensitivity, by varying actor prompt phrasing across replications and measuring the variance in decision distributions; and (iii) role adherence, verifying that agents do not access information outside their designated visibility tier (e.g., a provider citing ground-truth capability values that are structurally

hidden). These checks have no direct precedent in the classical ABM literature and constitute a novel validation requirement for LLM-in-the-loop simulations.

Ablation as structural diagnostic. The factorial ablation design in Section 6 doubles as a structural diagnostic: by removing individual actor types or feedback loops (e.g., no media, no policymaker, no funders), we test whether observed dynamics are emergent properties of multi-actor interaction or artifacts driven by a single component Sargent [2013]. If a finding (e.g., persistent gaming) vanishes when a specific actor is removed, it is attributable to that actor’s mechanism; if it persists across all ablations, it reflects a deeper structural property of the ecosystem.

6 Experiments and Results

We use the simulation as a hypothesis generator: by systematically removing ecosystem components and varying regulatory regimes, we identify which structural conditions appear to enable or suppress specific failure modes. The patterns described below are emergent properties of the simulated system and should be understood as candidate dynamics that motivate further empirical investigation, not as empirical claims about real-world AI markets.

We run 30-round experiments under eleven conditions: eight structural ablations that each remove one ecosystem component from the full baseline, and three regulatory-regime variants of the full ecosystem (Table 2).

Table 2: Experimental conditions. Ablations remove a single component from the full ecosystem under the balanced regulatory preset. Regulatory variants modify the policymaker’s intervention threshold, fine multiplier, and sanction duration.

Condition	Description
<i>Structural ablations (balanced preset)</i>	
No Media	Media actor removed; no public reporting on gaming
No Incidents	Safety-incident generation disabled
No Startups	No new entrants; incumbent-only market
No OpenCore	Open-source provider removed
No Funders	Venture-capital actor removed
No Bench. Evol.	Benchmarks never retire or update
Single Benchmark	One benchmark active throughout
Eval As Company	Evaluator charges providers for premium submissions
<i>Regulatory-regime variants (full ecosystem)</i>	
Balanced	Default preset (threshold 0.55, fine mult. 0.20, 3-round sanctions)
US Light-Touch	High threshold (0.75), low fines (0.10), short sanctions (2 rounds)
EU Precautionary	Low threshold (0.35), high fines (0.35), long sanctions (4 rounds)

All actor decisions are generated by Claude 3.5 Sonnet (`claude-3-5-sonnet-20241022`) at temperature 0.7 with a 1024-token completion limit. The simulation begins with five providers and four benchmark domains (coding, reasoning, math, safety); up to four additional

domains are introduced over the run with a six-round cooldown between introductions. Each condition uses a single seed (seed = 1); multi-seed replications for statistical inference are reported separately. Consumer agents (10 individual segments and 3 organizational segments) use heuristic decision rules; organizational consumers use LLM-based planning. Full hyperparameter values are listed in Table 1.

We report four principal findings, each stated as a hypothesis suggested by the simulation rather than an empirical claim.

Score inflation is persistent and regime-independent. Across all eleven conditions, aggregate score inflation (mean published score minus true capability) rises sharply in the first 5–10 rounds and stabilises at approximately 0.15–0.18 by round 30 (Figure 3). The US light-touch condition sustains the highest plateau; removing OpenCore reduces inflation modestly. The EU precautionary condition does not eliminate inflation but achieves a meaningfully better late-period validity–inflation tradeoff than the US condition (Figure 4). This pattern suggests that the incentive to game benchmarks may be sufficiently strong that regulatory escalation alone is unlikely to counteract it without structural changes to how evaluation signals map to competitive outcomes.

Aggregate Score Inflation Over Time
(mean score – true capability across all benchmarks and providers)

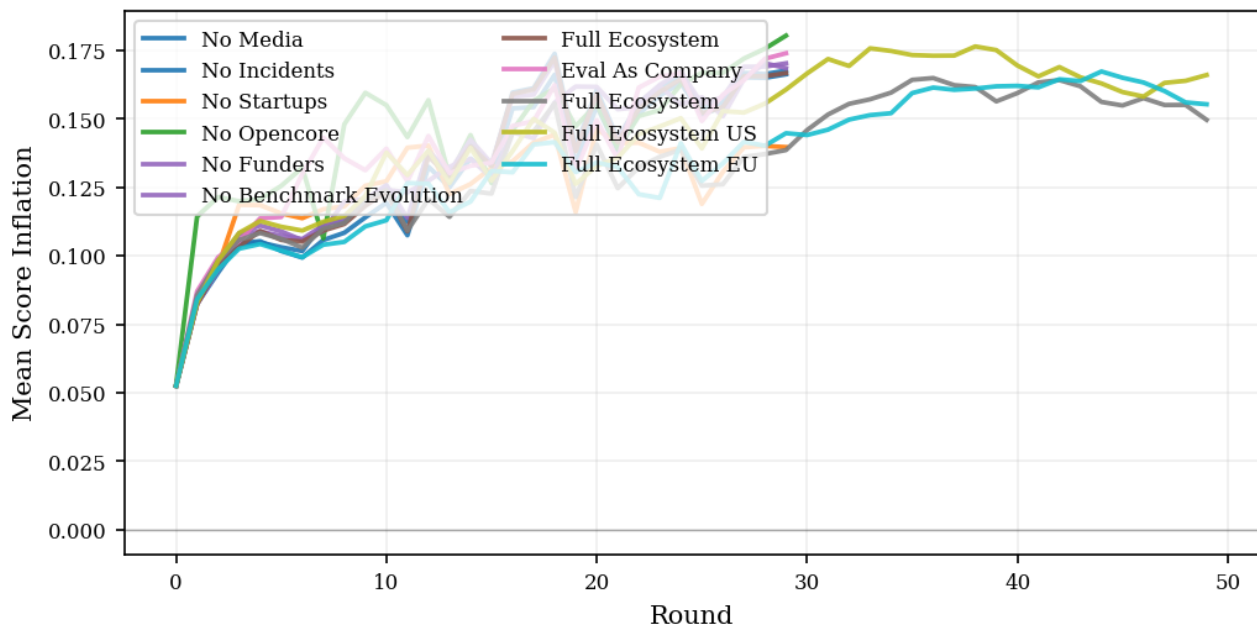


Figure 3: Score inflation trajectories across all eleven experimental conditions over 30 rounds. Inflation (published score minus true capability) rises sharply in early rounds and stabilises, with the US light-touch condition sustaining the highest plateau.

Late-Period Validity vs. Inflation Tradeoff
(each point = one experiment; last 10 rounds; ideal = upper-left)

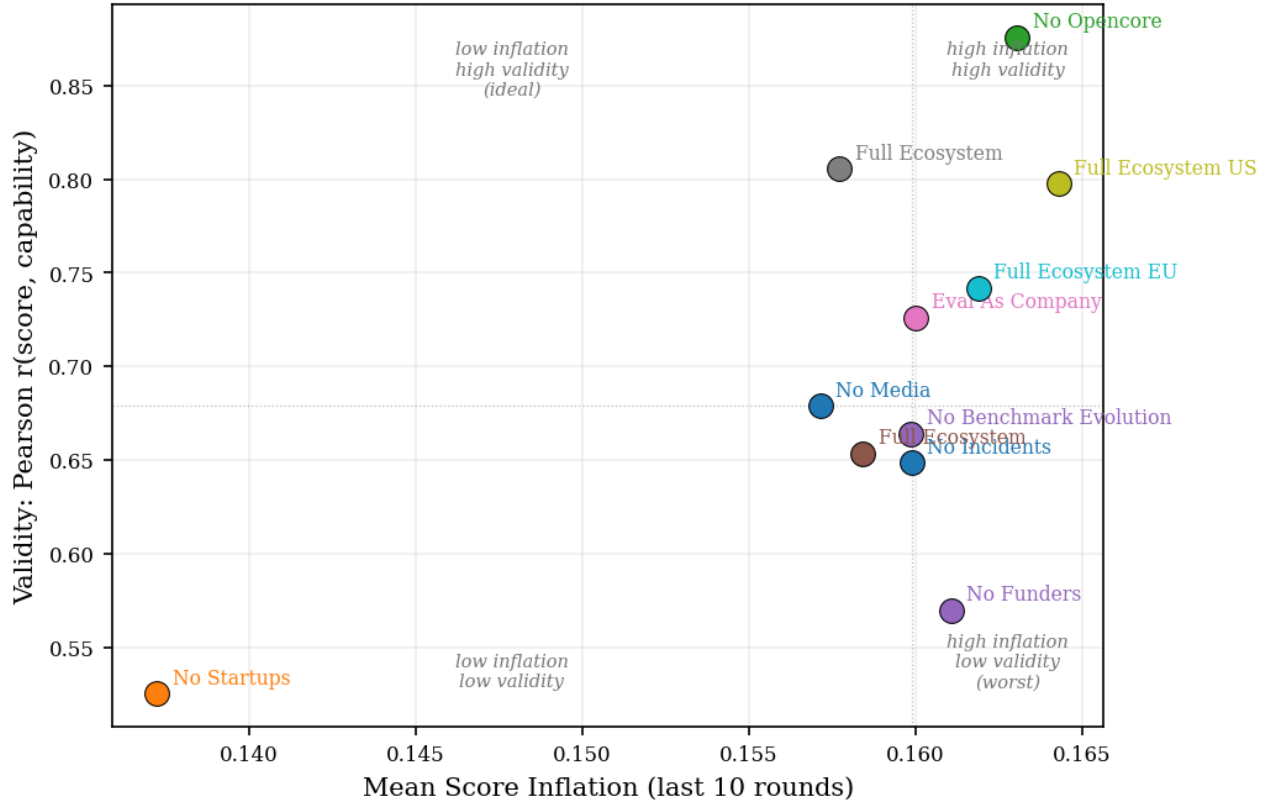


Figure 4: Late-period (rounds 20–30) validity vs. inflation scatterplot across conditions. Removing OpenCore achieves the best tradeoff (higher validity, moderate inflation); No Funders produces the worst validity outcome.

Gaming patterns are provider- and domain-specific. The gaming fingerprint heatmaps (Figure 5) show that providers with high evaluation engineering investment—Spark AI (45%) and OpenCore (35%)—exhibit the largest inflation signatures across most domains, while the safety-focused Apex AI shows the most muted inflation. Finance and Legal benchmarks display the largest condition-driven variance, consistent with their higher exploitability in the full-ecosystem runs (Figure 9). This differentiation across providers and domains is consistent with the hypothesis that gaming pressure concentrates where returns to evaluation engineering are highest and external auditing is hardest.

Benchmark validity partially recovers but does not hold. Rolling validity (Figure 6) frequently dips sharply as gaming pressure accumulates, then partially recovers when benchmarks retire and are replaced. The late-period validity–inflation scatterplot (Figure 4) shows that removing OpenCore achieves the best tradeoff (higher validity, moderate inflation),

consistent with OpenCore’s contamination multiplier accelerating Goodhart decay ecosystem-wide even though it receives no VC funding. This suggests that open-source providers may function as a structural accelerant of benchmark degradation even when excluded from the commercial incentive structures that otherwise drive gaming.

Gaming manifests primarily as floor bias, not amplification. The slope–intercept decomposition (Figure 7) shows that most benchmarks accumulate large positive intercepts (floor bias: a persistent score bonus shared by all providers) rather than slopes above 1.0 (amplification: higher-capability providers disproportionately rewarded). This is corroborated by the slope-vs-exploitability scatter (Figure 8), which shows no positive association between exploitability and slope across conditions. If this pattern were to hold in real systems, it would imply that gaming raises the entire leaderboard approximately uniformly, obscuring relative capability differences without systematically reordering providers—a failure mode that is harder to detect from leaderboard data alone.

Regulatory regime moderates but does not eliminate gaming. The EU precautionary preset achieves modestly higher late-period validity than the US light-touch preset (Figure 4), consistent with its earlier intervention threshold and more frequent escalations compressing the gaming gap. However, both regimes sustain comparable steady-state inflation levels, indicating that graduated regulatory escalation alone may be insufficient to counteract the underlying market incentive to game benchmarks when evaluation signals directly drive capital allocation.

6.1 Discussion

The simulation produces four structural patterns with relevance to how the evaluation ecosystem is designed. We state these as hypotheses suggested by the simulation rather than empirical findings.

- **Gaming as a stable equilibrium (hypothesis).** Score inflation emerged rapidly and persisted across all conditions, including those with active benchmark evolution and regulatory intervention. This suggests that gaming pressure may be a stable property of competitive evaluation markets rather than a correctable artifact, and that no single ecosystem component—when removed—was sufficient to eliminate it.
- **Floor bias as a detection challenge (hypothesis).** Because the simulated gaming raised all providers’ scores rather than selectively rewarding frontrunners, neither consumers nor funders in the simulation could reliably distinguish capability leaders from skilled benchmark optimizers using leaderboard data alone. If this floor-bias pattern generalizes, it would suggest that relative rankings on public leaderboards may be more stable than their absolute values imply, but for the wrong reasons.
- **Open-source providers as contamination accelerants (hypothesis).** OpenCore’s $1.8\times$ contamination multiplier accelerated Goodhart decay ecosystem-wide while remaining exempt from sanctions. Removing it produced the single largest improvement

in validity across all ablations. This points to a structural asymmetry: open-source release may export benchmark degradation pressure to the broader ecosystem even when the releasing actor is not subject to competitive or regulatory incentives.

- **Capital concentration and gaming incentives (hypothesis).** The *No Funders* condition produced the worst validity outcome (Figure 4), suggesting a counterintuitive dynamic: removing venture capital did not reduce gaming but may have eliminated a moderating signal (gaming suspicion adjustments) that funders otherwise introduce. This warrants closer examination of how capital allocation interacts with evaluation incentives in practice.

Each ablation reveals which structural conditions the simulation associates with enabling or suppressing specific failure modes, providing a basis for reasoning about where systemic interventions might be most effective and what empirical evidence would be needed to evaluate such interventions in real ecosystems.

6.2 Simulation Diagnostics

[Sang: WIP – Place holder for now!]

We apply the diagnostic protocol described in Section 5 to all eleven conditions. Full results (per-experiment plots, drift heatmaps, ablation effect sizes) appear in Appendix 10.1; we summarize the key findings here.

Pattern-oriented validation. Table 3 reports pass rates for seven qualitative patterns across all conditions. Score inflation and regulatory escalation hold universally (11/11). Benchmark turnover fails only when benchmark evolution is structurally disabled (Single Benchmark, No Bench. Evolution)—expected by design. Commoditization shock fails only under No OpenCore, confirming that this pattern is attributable to open-source dynamics. The Funding Follows Scores pattern fails in 9 of 11 conditions: funders in the simulation correlate their allocations more with inferred capability than with published scores, suggesting that the funder model partially “sees through” benchmark gaming.

Table 3: Pattern-oriented validation results across eleven conditions. Each cell indicates whether the qualitative pattern was observed. Patterns are evaluated on single-seed runs; multi-seed confirmation is pending.

Pattern	Pass Rate	Notable Failures
Score Inflation	11/11	—
Regulatory Escalation	11/11	—
Benchmark Turnover	9/11	Single Bench., No Bench. Evol.
Commoditization Shock	10/11	No OpenCore
Safety Incident Response	10/11	No Funders
Gaming Persistence	8/11	No Startups, 50-round runs
Funding Follows Scores	2/11	Most conditions

Behavioral coherence. Strategy drift, measured as the mean cosine distance between consecutive round allocations, remains below 0.02 for all providers across all conditions, indicating that LLM actors maintain stable strategic profiles rather than oscillating erratically. Role adherence checks flag 52–225 term-level violations per experiment (e.g., actors using phrases like “true capability” in reasoning traces). Manual inspection suggests these reflect LLM reasoning about the *concept* of unobserved quality rather than actual information leakage, since the ground-truth values are structurally excluded from actor prompts. Full drift heatmaps and violation counts are reported in Appendix 10.1.

6.3 Limitations and Complementary Methods

Our LLM-powered agent-based simulation is one of several methodologies that could be applied to study the AI evaluation ecosystem. Each alternative captures dynamics that our approach handles only partially, and we view these methods as complements rather than competitors.

Our use of a stylized simulation follows a well-established methodological tradition. *Ie et al.* [2019] articulate the rationale clearly in the context of recommender systems: “Our goal is not to create a ‘perfect’ simulator; we do not expect policies learned in simulation to be deployed in live systems. Rather, we expect simulations that mirror specific aspects of user behavior found in real systems to serve as a controlled environment for developing, evaluating and comparing [...] models and algorithms.” The same logic applies here. Our simulation does not predict the trajectory of real evaluation markets; it provides a controlled environment in which structural hypotheses—about feedback loops, incentive misalignment, and regulatory leverage—can be tested counterfactually. Static observational data cannot support such counterfactual reasoning because the key quantities (true capability, evaluation-engineering investment, benchmark validity) are not directly observable and the structural interventions we study (removing funders, disabling benchmark evolution) have no real-world analogue to exploit as natural experiments. Simulation fills this gap by enabling controlled ablation at the cost of ecological fidelity—a tradeoff that is explicit by design.

Game-theoretic analysis. The investment-allocation game at the core of our simulation—providers choosing between R&D, evaluation engineering, and safety under competitive pressure—admits game-theoretic formalization. A mechanism-design analysis could characterize whether gaming constitutes a Nash equilibrium under stated payoff assumptions and, more importantly, identify scoring rules or audit mechanisms that make gaming incentive-incompatible. Our simulation observes that gaming is persistent across all conditions but cannot *prove* it is an equilibrium; game-theoretic grounding would strengthen this claim. The limitation of purely analytic approaches is tractability: the multi-actor, multi-benchmark, multi-round structure with graduated regulatory escalation, benchmark retirement, and endogenous market entry resists closed-form solution.

System dynamics. Stock-and-flow models represent aggregate quantities (total gaming investment, benchmark validity, consumer trust) as continuous variables connected by feedback equations. System dynamics excels at making feedback-loop architecture explicit and

identifying leverage points Meadows [2008], and could complement our work by characterizing long-run equilibria and tipping points in the evaluation ecosystem. However, system dynamics assumes homogeneous populations and cannot represent the provider-level heterogeneity (e.g., safety-focused Apex AI versus gaming-heavy Spark AI) or domain-specific effects (e.g., Finance and Legal benchmarks exhibiting higher exploitability) that our simulation reveals as important drivers of ecosystem dynamics.

Empirical and econometric methods. Observational data—longitudinal benchmark scores, model release timelines, funding events, adoption metrics—could validate or refute the hypotheses our simulation generates. Natural experiments such as benchmark transitions (e.g., MMLU to MMLU-Pro) offer quasi-experimental designs for measuring Goodhart decay in practice. The principal limitation is data availability: quantities central to our analysis (true capability, evaluation-engineering investment, benchmark validity decay rates) are not directly observable. Pairing simulation-generated hypotheses with empirical tests where data permits is the most promising path to external validation.

Expert elicitation. Delphi studies or structured scenario-planning exercises with AI researchers, benchmark developers, and policymakers could calibrate simulation parameters and validate whether the emergent dynamics we observe are consistent with practitioner experience. Expert judgment captures tacit institutional knowledge (political constraints, lobbying dynamics, informal norms) that our simulation does not model. However, experts cannot reliably trace multi-step feedback chains—the counterintuitive finding that removing funders *worsens* validity, for instance, is precisely the kind of emergent dynamic that eludes unaided human reasoning.

Scope and external validity. Beyond methodological alternatives, we note several scope limitations. The simulation uses a single LLM (Claude 3.5 Sonnet) for all strategic actors; cross-model replication would test whether findings are robust to the choice of backbone model. Consumer agents use heuristic decision rules rather than LLM-based reasoning, which may understate demand-side strategic behavior. All results are based on single-seed runs; multi-seed replications with confidence intervals are needed before any finding can be treated as statistically reliable. Finally, the simulation models a stylized ecosystem with five providers and parameterized benchmarks; the mapping to real-world markets with hundreds of models and evolving evaluation infrastructure requires careful qualification.

7 Research Agenda: What the Simulation Can and Cannot Do

The simulation is an instrument, not an oracle. Its value lies in the range of counterfactual questions it can pose under controlled conditions—and equally in recognizing which questions it cannot yet address, which define both the extension roadmap and the empirical work that must complement simulation-based reasoning. We organize this section around concrete

research and policy questions, grouped by what the current framework enables, what it does not yet model but could, and what requires fundamentally different evidence.

7.1 Questions the Simulation Can Explore

Regulatory design and calibration. Because the policymaker actor is parameterized by intervention threshold, fine structure, and escalation ladder, the simulation can serve as a testbed for regulatory design questions that are difficult to study empirically.

- *What is the marginal return to lowering intervention thresholds?* The EU–US comparison in Section 6 suggests diminishing returns to stricter regulation, but a finer-grained sweep over threshold values could identify inflection points where additional regulatory stringency produces disproportionate gains—or ceases to matter.
- *Do disclosure mandates reduce gaming?* A condition requiring providers to publish their evaluation-engineering allocation (converting it from private to public state) would test whether transparency deters gaming or merely shifts it to less observable channels.
- *Should open-source providers face evaluation-specific regulation?* OpenCore’s exemption from sanctions is a structural feature of the current simulation. Ablating this exemption would clarify whether the contamination-accelerant effect identified in Section 6 can be mitigated by extending regulatory coverage to open-weight releases.
- *How does regulatory harmonization compare to fragmentation?* Extending the simulation to multiple jurisdictions with providers choosing where to submit evaluations would model regulatory arbitrage—a dynamic that single-jurisdiction experiments cannot capture.

Evaluation infrastructure design. The benchmark lifecycle model—with its validity decay, exploitability growth, and retirement mechanics—enables systematic study of how evaluation institutions might be structured.

- *How frequently should benchmarks be refreshed?* Varying the retirement threshold and degradation rate can identify the rotation frequency at which validity is maintained without imposing unsustainable development costs on evaluation providers.
- *Does benchmark diversity protect against gaming?* The single-benchmark ablation already shows that concentrating evaluation on one domain worsens outcomes. Intermediate conditions (two, four, or sixteen domains) could quantify the marginal value of additional benchmark diversity.
- *What is the value of held-out evaluation?* Introducing a benchmark class that is invisible to providers until scoring (modeling private or held-out test sets) would test whether information asymmetry between evaluators and providers is a viable structural defense against gaming.

- *Can evaluator incentives be restructured?* The Eval-As-Company ablation begins to explore this, but further conditions—such as evaluators paid by consumers rather than providers, or evaluators rewarded for predictive validity—would map the design space for evaluation business models.

Market structure and competition. The consumer, funder, and startup-entry mechanics allow the simulation to explore how competitive dynamics interact with evaluation integrity.

- *At what market concentration does gaming become self-reinforcing?* Sweeping over the number of initial providers or consumer switching friction can identify concentration thresholds beyond which gaming locks in—information relevant to antitrust considerations specific to AI markets.
- *Can mission-driven funding shift the equilibrium?* Increasing the share of foundation or government funding relative to VC, or introducing quality-conditional funding rules, would test whether capital allocation is a viable policy lever for evaluation integrity.
- *How do barriers to entry interact with gaming incentives?* The startup-entry model provides a natural framework for studying whether lowering entry barriers (e.g., through compute subsidies or shared evaluation infrastructure) reduces gaming by intensifying competition, or amplifies it by adding more actors competing on benchmarks.

Consumer and information dynamics. The 39-segment consumer market and media actor enable questions about how information flows shape downstream outcomes.

- *Does consumer sophistication reduce gaming payoffs?* Shifting the distribution of consumer archetypes toward lower leaderboard trust (more experience-driven or cautious consumers) tests whether demand-side pressure can substitute for supply-side regulation.
- *What role does media coverage play in equilibrium?* The no-media ablation shows media’s absence matters, but more nuanced conditions—varying media accuracy, publication delay, or sensationalism—could identify which properties of information intermediaries are most consequential.

7.2 Current Limitations and Extension Roadmap

One limitation is intrinsic to the simulation-as-instrument approach; the rest are gaps in the current implementation that define a natural extension roadmap.

Intrinsic: qualitative hypotheses, not point predictions. The simulation generates comparative and structural hypotheses, not quantitative forecasts. Specific numbers—market shares, inflation magnitudes, incident rates—are properties of the parameterization, not of real markets. The simulation can suggest that light-touch regulation is associated with higher gaming *relative to* precautionary regulation; it cannot predict that gaming will reach 18% under a specific regime. This is not a gap to be closed by better engineering—it reflects the epistemic status of any simulation whose parameters are not fully calibrated to empirical data.

The appropriate use is to identify *which structural conditions matter most*, then prioritize empirical measurement of those conditions.

The remaining items describe dynamics that the framework does not yet model but that its architecture can accommodate. We flag each as an extension opportunity rather than a fixed boundary.

Disaggregated gaming mechanisms. Evaluation engineering is currently modeled as a single aggregate investment that inflates scores via the βe term. In practice, gaming encompasses a heterogeneous set of strategies—data contamination, benchmark-specific fine-tuning, prompt engineering, selective checkpoint submission—each with different costs, detectability, and countermeasures Sainz et al. [2023], Balloccu et al. [2024]. Decomposing the evaluation-engineering channel into distinct strategy types, each with its own cost curve and detection probability, would allow the simulation to study which countermeasures (e.g., contamination detection, dynamic test sets) are most effective against which gaming vectors.

Richer capability dynamics. True capability currently evolves as a smooth function of investment with stochastic perturbation. This does not capture discontinuous breakthroughs, paradigm shifts (e.g., a new architecture class), or the specific mechanisms by which training compute translates into model performance. Incorporating threshold effects or S-curve capability trajectories—potentially informed by scaling-law research—would enable the simulation to study how the strategic landscape changes when capability gains become harder or easier to achieve, and whether gaming incentives intensify near capability plateaus.

Multi-jurisdictional dynamics. The current framework models a single regulatory jurisdiction. Real evaluation ecosystems span multiple regulatory regimes, and providers strategically choose where to operate, submit evaluations, and incorporate. Extending the simulation to multiple concurrent policymaker actors with jurisdiction-specific rules and provider mobility would enable study of regulatory arbitrage, jurisdictional competition, and the extraterritorial reach of regulations like the EU AI Act—first-order policy concerns that the single-jurisdiction setup cannot yet address.

Collusion, lobbying, and political economy. Providers currently act independently, interacting only through market competition and shared evaluation infrastructure. Adding communication channels between providers (enabling tacit coordination on benchmark standards), lobbying mechanics (providers investing to shift regulatory parameters), or revolving-door dynamics would open a class of political-economy questions—such as whether industry self-regulation converges on evaluation standards that favor incumbents—that may be among the most important determinants of real-world evaluation governance.

Compute, data, and infrastructure constraints. The simulation abstracts away the physical infrastructure of AI development. Introducing resource constraints—GPU budgets, training data access, energy costs—as an additional dimension of provider strategy would allow the simulation to study whether compute scarcity changes the relative returns to gaming

versus fundamental research, and whether infrastructure subsidies function as an indirect lever on evaluation integrity.

Temporal calibration. A simulation “round” does not currently correspond to a fixed real-world time period. Calibrating round duration to observable evaluation cycles—major benchmark releases, leaderboard update frequencies, or regulatory review periods—would enable the simulation to generate temporally grounded predictions (e.g., “under these conditions, benchmark validity degrades within two release cycles”) rather than purely ordinal comparisons.

Endogenous evaluation criteria. The simulation currently takes the set of benchmark domains and the structure of the scoring model as exogenous. In practice, what counts as a good benchmark, which capabilities are deemed important, and how evaluation standards evolve are themselves products of the ecosystem dynamics we study. Allowing actors to propose, contest, and adopt new evaluation criteria—making the definition of “capability” itself a strategic variable—would capture a feedback loop that the current framework treats as fixed background.

7.3 Bridging Simulation and Empirical Evidence

The gap between what the simulation currently does and what it could do with targeted extensions defines a natural research program. Three directions appear most productive for closing this gap.

Empirical calibration. Where longitudinal data on benchmark performance, market shares, and evaluation practices exist—even imperfectly—they can be used to constrain simulation parameters and narrow the space of plausible dynamics. Public leaderboard histories (e.g., from HELM Liang et al. [2023] or Chatbot Arena Chiang et al. [2024]), funding disclosures, and regulatory action databases provide partial but growing empirical grounding. Calibration is also the prerequisite for temporal grounding: mapping simulation rounds to real-world timescales requires observable anchor points.

Natural experiments. Real-world policy changes—the EU AI Act’s phased implementation, China’s algorithmic recommendation regulations, or individual platform decisions to change evaluation methodology—create natural experiments whose observed effects can be compared against simulation predictions under matched conditions. Discrepancies between simulated and observed dynamics would identify where the model’s structural assumptions need revision and which extensions on the roadmap above should be prioritized.

Mechanism-specific empirical studies. Each extension opportunity listed above is paired with an empirical gap that constrains how it should be built. Contamination detection studies Sainz et al. [2023], Balloccu et al. [2024] would inform the disaggregation of gaming mechanisms. Surveys of enterprise procurement decisions would ground richer consumer models. Case studies of regulatory lobbying in AI governance would parameterize the political-economy extension. The simulation’s value is greatest when its extensions are disciplined by empirical work that tests structural assumptions one at a time, rather than attempting to validate the entire model against a single aggregate outcome.

8 Complementary Methods

Our LLM-powered agent-based simulation is one of several methodologies that could be applied to study the AI evaluation ecosystem. Each alternative captures dynamics that our approach handles only partially, and we view these methods as complements rather than competitors.

Game-theoretic analysis. The investment-allocation game at the core of our simulation—providers choosing between R&D, evaluation engineering, and safety under competitive pressure—admits game-theoretic formalization. A mechanism-design analysis could characterize whether gaming constitutes a Nash equilibrium under stated payoff assumptions and, more importantly, identify scoring rules or audit mechanisms that make gaming incentive-incompatible. Our simulation observes that gaming is persistent across all conditions but cannot *prove* it is an equilibrium; game-theoretic grounding would strengthen this claim. The limitation of purely analytic approaches is tractability: the multi-actor, multi-benchmark, multi-round structure with graduated regulatory escalation, benchmark retirement, and endogenous market entry resists closed-form solution.

System dynamics. Stock-and-flow models represent aggregate quantities (total gaming investment, benchmark validity, consumer trust) as continuous variables connected by feedback equations. System dynamics excels at making feedback-loop architecture explicit and identifying leverage points Meadows [2008], and could complement our work by characterizing long-run equilibria and tipping points in the evaluation ecosystem. However, system dynamics assumes homogeneous populations and cannot represent the provider-level heterogeneity (e.g., safety-focused Apex AI versus gaming-heavy Spark AI) or domain-specific effects (e.g., Finance and Legal benchmarks exhibiting higher exploitability) that our simulation reveals as important drivers of ecosystem dynamics.

Empirical and econometric methods. Observational data—longitudinal benchmark scores, model release timelines, funding events, adoption metrics—could validate or refute the hypotheses our simulation generates. Natural experiments such as benchmark transitions (e.g., MMLU to MMLU-Pro) offer quasi-experimental designs for measuring Goodhart decay in practice. The principal limitation is data availability: quantities central to our analysis (true capability, evaluation-engineering investment, benchmark validity decay rates) are not directly observable. Pairing simulation-generated hypotheses with empirical tests where data permits is the most promising path to external validation.

Expert elicitation. Delphi studies or structured scenario-planning exercises with AI researchers, benchmark developers, and policymakers could calibrate simulation parameters and validate whether the emergent dynamics we observe are consistent with practitioner experience. Expert judgment captures tacit institutional knowledge (political constraints, lobbying dynamics, informal norms) that our simulation does not model. However, experts cannot reliably trace multi-step feedback chains—the counterintuitive finding that removing

fundamentals *worsens* validity, for instance, is precisely the kind of emergent dynamic that eludes unaided human reasoning.

Scope and external validity. Beyond methodological alternatives, we note several scope limitations. The simulation uses a single LLM (Claude 3.5 Sonnet) for all strategic actors; cross-model replication would test whether findings are robust to the choice of backbone model. Consumer agents use heuristic decision rules rather than LLM-based reasoning, which may understate demand-side strategic behavior. All results are based on single-seed runs; multi-seed replications with confidence intervals are needed before any finding can be treated as statistically reliable. Finally, the simulation models a stylized ecosystem with five providers and parameterized benchmarks; the mapping to real-world markets with hundreds of models and evolving evaluation infrastructure requires careful qualification.

9 Conclusion

This paper has argued that AI evaluation is best understood not as a technical measurement problem but as a system of interacting actors whose incentives, information, and strategic behavior jointly determine what gets measured, how results are used, and which failure modes emerge. We developed a systems-level framework mapping the actors, signals, and feedback loops that constitute the evaluation ecosystem, and instantiated it as an agent-based simulation that serves as a hypothesis generator for ecosystem dynamics.

The simulation demonstrates the value of this approach. By ablating individual ecosystem components and varying regulatory regimes, we identified structural pathways—such as open-source contamination pressure, capital concentration, and floor bias—that would be difficult to anticipate from studying any single benchmark or actor in isolation. These findings are descriptive rather than prescriptive: they characterize *how* systemic failures emerge from actor interactions, generating hypotheses that can guide future empirical investigation.

This matters because the alternative tools for understanding evaluation dynamics are severely limited. Policy interventions in the evaluation ecosystem are difficult to study through conventional means. Randomized controlled trials are expensive, slow, and in many cases impossible—one cannot randomly assign regulatory regimes to national AI ecosystems or experimentally withhold funding from evaluation institutions. Natural experiments are rare and confounded. Observational studies of evaluation practices face the usual challenges of causal identification, compounded by the strategic behavior of actors who anticipate being observed. As a result, policymakers and evaluation designers often operate with limited empirical grounding, relying on analogies to other domains or on intuitions about how actors will respond to incentive changes.

Agent-based simulation offers a complementary path forward. A simulation grounded in a well-specified ecosystem model can explore counterfactuals that are infeasible to test empirically: What happens if open-source model weights accelerate benchmark contamination? How does the presence or absence of a media actor change the speed of regulatory response? Does mission-driven funding meaningfully counteract venture capital concentration? These are precisely the kinds of questions that matter for evaluation governance but resist traditional empirical methods. Simulation does not replace empirical evidence, but it can narrow the

space of plausible hypotheses, identify which structural conditions deserve the most scrutiny, and reveal non-obvious interaction effects between ecosystem components.

The framework and simulation we present are a first step. Several directions for future work follow naturally. First, the simulation can be extended to model additional actors and feedback channels—for instance, media dynamics in greater detail, or the role of open-source model ecosystems in shaping evaluation norms. Second, calibrating simulation parameters against empirical data from real evaluation platforms would strengthen the connection between simulated and observed dynamics. Third, the ablation methodology can be applied to evaluate proposed interventions before deployment: testing whether a new disclosure requirement or funding mechanism achieves its intended effect within the simulated ecosystem, and under what conditions it fails. Finally, the systems framework itself can inform the design of evaluation institutions that are robust to the strategic pressures we have identified, by making explicit the feedback loops that current designs leave unaddressed.

More broadly, we hope this work contributes to establishing evaluation as a domain that warrants its own systems-level science—one that takes seriously the fact that the act of measuring AI systems changes the systems being measured, the actors who build them, and the institutions that govern them.

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10 Appendix

10.1 Simulation Diagnostic Results

This appendix reports full diagnostic results for all eleven experimental conditions described in Section 6. All diagnostics follow the protocol specified in Section 5. Results are based on single-seed runs (seed = 1); multi-seed replications with confidence intervals are planned as a follow-up.

10.1.1 Pattern-Oriented Validation (Full Matrix)

Table 4 reports the pass/fail status of each pattern for every experiment. The fifteen rows include duplicate runs (exp_003 duplicates exp_001) and 50-round extensions (exp_013–015) of the 30-round baselines.

Interpretation. Score Inflation and Regulatory Escalation pass universally, confirming that these are structural properties of the simulation rather than artifacts of specific actor configurations. Benchmark Turnover fails only when benchmark evolution is disabled by design (Single Benchmark, No Bench. Evolution). Commoditization Shock fails only under No OpenCore, confirming the pattern is driven by open-source dynamics. Gaming Persistence fails in the 50-round runs and No Startups, where evaluation engineering investment declines in later rounds as providers shift toward other strategies—a finding that itself merits further investigation. The Funding Follows Scores pattern fails in 12 of 14 runs (excluding Single Benchmark and No Bench. Evolution, where it passes), indicating that funders in the

Table 4: Full pattern validation matrix across all fifteen experiments. ✓ = pattern observed; × = pattern not observed. Shaded cells indicate expected failures due to the ablation design.

Condition	Score Infl.	Bench. Turn.	Reg. Escal.	Commod. Shock	Safety Resp.	Gaming Persist.	Fund. → Scores
Full Eco. (Balanced, 30r)	✓	✓	✓	✓	✓	✓	×
Full Eco. (US, 30r)	✓	✓	✓	✓	✓	✓	×
Full Eco. (EU, 30r)	✓	✓	✓	✓	✓	✓	×
Full Eco. (Balanced, 50r)	✓	✓	✓	✓	✓	×	×
Full Eco. (US, 50r)	✓	✓	✓	✓	✓	×	×
Full Eco. (EU, 50r)	✓	✓	✓	✓	✓	×	×
No Media	✓	✓	✓	✓	✓	✓	×
No Incidents	✓	✓	✓	✓	✓	✓	×
No Startups	✓	✓	✓	✓	✓	×	×
No OpenCore	✓	✓	✓	×	✓	✓	×
Single Benchmark	✓	×	✓	✓	✓	✓	✓
No Funders	✓	✓	✓	✓	×	✓	×
No Bench. Evol.	✓	×	✓	✓	✓	✓	×
Eval As Company	✓	✓	✓	✓	✓	✓	×

simulation correlate allocations more with inferred capability than published benchmark scores.

10.1.2 Ablation Effect Sizes

Table 5 reports the difference in key metrics between each ablation and the balanced full-ecosystem baseline (exp_011), averaged over the last five rounds.

Table 5: Ablation effect sizes relative to the balanced full-ecosystem baseline (exp_011), averaged over rounds 26–30. Bold values indicate the largest absolute effect for each metric.

Ablation	Δ Gaming	Δ Capability	Δ Eval Eng.	Δ HHI	Δ Satisfaction
No Startups	−0.027	+0.073	− 0.177	+0.172	+0.026
No OpenCore	+0.012	−0.068	+ 0.142	+0.114	+0.052
Single Bench.	−0.018	−0.002	+0.005	+ 0.292	+0.085
No Bench. Evol.	+0.002	−0.004	+0.005	+0.219	−0.015
No Funders	+0.004	−0.028	+0.008	−0.026	+0.013
No Media	−0.000	−0.005	−0.000	+0.193	+0.054
No Incidents	+0.001	−0.001	+0.006	−0.011	+0.001
Eval As Company	+0.003	+0.004	+0.020	+0.126	+0.045

Key observations. Removing startups produces the largest reduction in evaluation engineering (−0.177), suggesting that competitive entry pressure is a primary driver of gaming investment. Conversely, removing OpenCore increases evaluation engineering (+0.142), consistent with the hypothesis that open-source contamination reduces the marginal return

to gaming for closed providers. Single Benchmark produces the largest increase in market concentration (HHI +0.292), indicating that benchmark diversity acts as a structural counterweight to market consolidation.

10.1.3 Behavioral Coherence

Strategy drift. Table 6 reports the mean cosine distance between consecutive round-to-round investment allocations for each incumbent provider, averaged across all eleven conditions. Values below 0.02 indicate stable strategic behavior.

Table 6: Mean strategy drift (cosine distance) per provider, averaged across all experimental conditions. All values remain below 0.02, indicating stable LLM actor behavior.

Provider	Mean Drift
Orion Labs	0.0080
Apex AI	0.0091
Genesis Systems	0.0089
Mirage AI	0.0107
OpenCore	0.0110

Role adherence. All fifteen experiments exhibit between 52 and 225 term-level violations in actor reasoning traces (Table 7). Violations are flagged when actors use phrases such as “true capability” or “actual capability”—terms that reference ground-truth state structurally excluded from their prompts.

Table 7: Role adherence violations per experiment. Violations count instances where actor traces reference ground-truth information that is structurally excluded from their prompts.

Condition	Violations
Full Eco. (Balanced, 30r)	107
Full Eco. (US, 30r)	82
Full Eco. (EU, 30r)	87
Full Eco. (Balanced, 50r)	206
Full Eco. (US, 50r)	225
Full Eco. (EU, 50r)	182
No Media	74
No Incidents	83
No Startups	83
No OpenCore	52
Single Benchmark	98
No Funders	70
No Bench. Evol.	67
Eval As Company	74

Manual inspection of flagged traces indicates that actors reason about the *concept* of unobserved quality (e.g., “our true capability may be lower than our published score suggests”) rather than accessing actual ground-truth values. The 50-round experiments accumulate more violations proportionally (≈ 4 per round), consistent with a per-round hallucination rate rather than information leakage. Violations scale linearly with experiment length, supporting this interpretation.

Provider Gaming Fingerprint
(mean score – true capability per provider per benchmark)

No Media

Company	Performance Metrics (Q1-Q4)								
	Q1 Performance		Q2 Performance		Q3 Performance		Q4 Performance		
	coding	reasoning	math	safety	writing	medical	legal	finance	
	Orion Labs	+0.14	+0.10	+0.16	+0.11	+0.23	+0.15	+0.12	+0.05
	Apex AI	+0.12	+0.18	+0.09	+0.20	+0.17	+0.14	+0.02	+0.17
	Genesis Systems	+0.11	+0.14	+0.10	+0.09	+0.15	+0.11	+0.12	+0.07
	Mirage AI	+0.07	+0.12	+0.11	+0.09	+0.20	+0.13	+0.05	+0.07
	OpenCore	+0.14	+0.10	+0.10	+0.10	+0.21	+0.12	+0.05	-0.01
OneAI	+0.23	+0.31	+0.20	+0.18	+0.28	+0.24	+0.07	+0.20	

No Incidents

Company	Performance Metrics (Q1-Q2)								
	coding	reasoning	math	safety	writing	medical	legal	finance	
	Orion Labs	+0.16	+0.11	+0.17	+0.12	+0.25	+0.15	+0.12	+0.06
	Apex AI	+0.12	+0.17	+0.09	+0.18	+0.16	+0.12	+0.01	+0.17
	Genesis Systems	+0.11	+0.13	+0.09	+0.09	+0.15	+0.11	+0.11	+0.07
	Mirage AI	+0.09	+0.13	+0.13	+0.10	+0.21	+0.14	+0.07	+0.07
	OpenCore	+0.13	+0.10	+0.10	+0.10	+0.22	+0.12	+0.06	-0.01
	OneAI	+0.24	+0.31	+0.23	+0.18	+0.28	+0.23	+0.07	+0.21

No Startups

Company	Performance Metrics (Q1-Q2)							
	coding	reasoning	math	safety	writing	medical	legal	finance
Orion Labs	+0.13	+0.14	+0.14	+0.11	+0.16	+0.12	+0.18	+0.04
Apex AI	+0.12	+0.18	+0.08	+0.22	+0.15	+0.05	+0.04	+0.06
Genesis Systems	+0.11	+0.16	+0.15	+0.10	+0.13	+0.10	+0.05	+0.06
Mirage AI	+0.08	+0.14	+0.16	+0.10	+0.17	+0.05	+0.14	+0.05
OpenCore	+0.18	+0.14	+0.12	+0.08	+0.18	+0.13	+0.10	+0.15
OneAI								

No Opencore

Company	Performance Metrics (Q1-Q2)								
	Q1 Performance				Q2 Performance				
	coding	reasoning	math	safety	writing	medical	legal	finance	
	Orion Labs	+0.09	+0.16	+0.19	+0.09	+0.14	+0.19	+0.06	-0.01
	Apex AI	+0.14	+0.18	+0.14	+0.12	+0.16	+0.06	+0.14	+0.17
	Genesis Systems	+0.08	+0.13	+0.17	+0.10	+0.16	+0.08	+0.17	+0.10
	Mirage AI	+0.13	+0.18	+0.15	+0.06	+0.32	+0.12	+0.14	+0.14
OpenCore	Overall Average								
OneAI	+0.11	+0.20	+0.17	+0.14	+0.30	+0.18	+0.15	+0.24	

Single Benchmark

Company	Project	Phase	Progress (%)
Orion Labs	Apex AI	Phase 1	100
		Phase 2	85
Genesis Systems	Mirage AI	Phase 1	70
		Phase 2	40

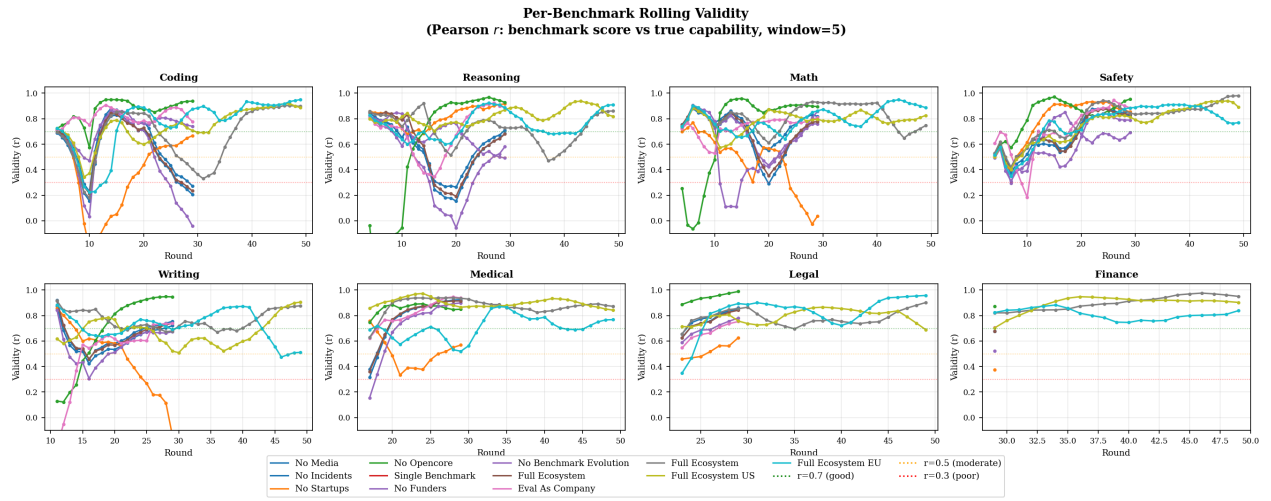


Figure 6: Rolling benchmark validity (α) over time across conditions. Validity dips under gaming pressure and partially recovers at benchmark retirement, but does not return to initial levels.

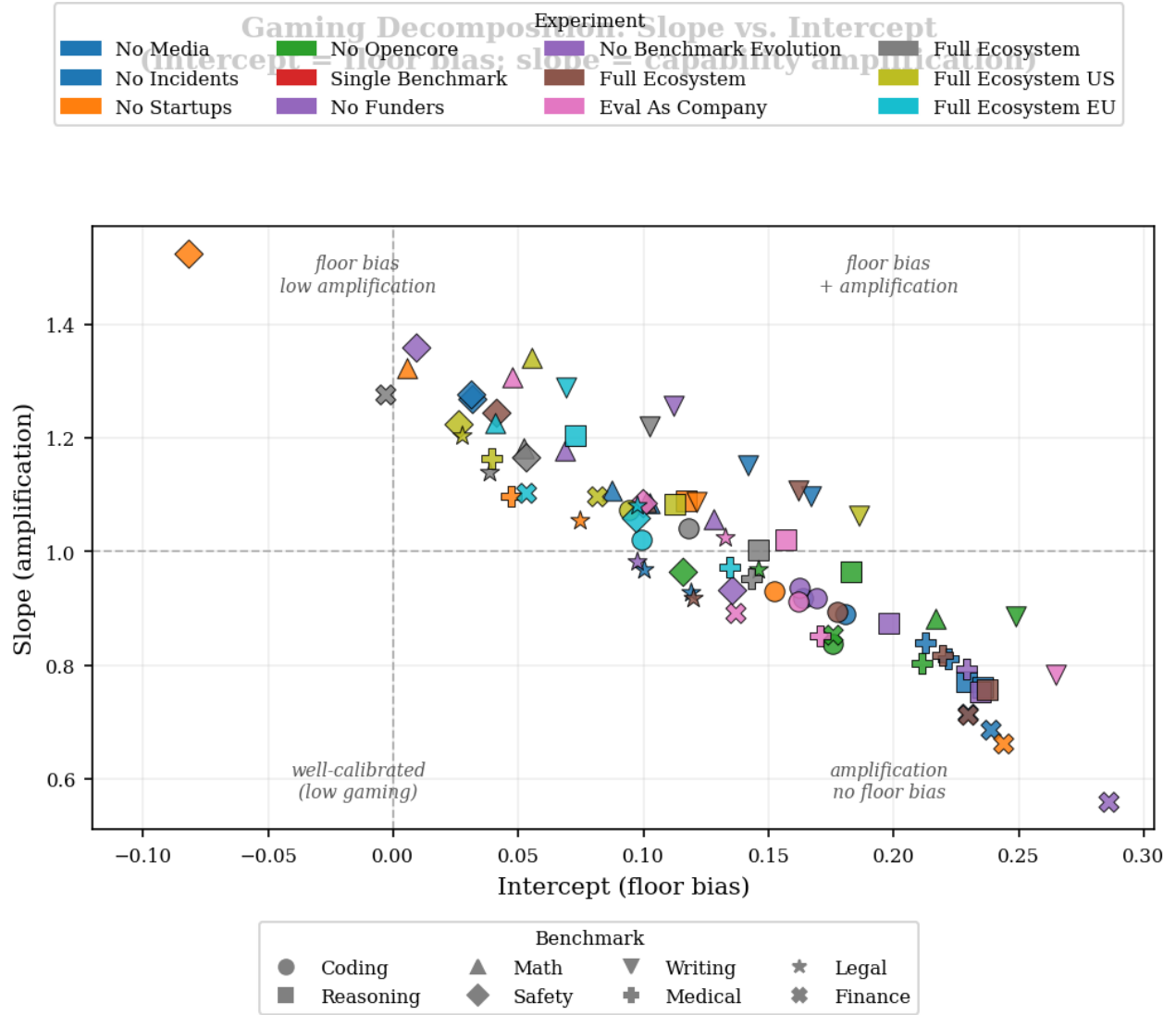


Figure 7: Slope–intercept decomposition of the published-score vs. true-capability regression across benchmarks and conditions. Large positive intercepts (floor bias) dominate over amplification (slope > 1).

OLS Slope vs. Benchmark Exploitability
(tests whether high-exploitability benchmarks decouple scores from capability)

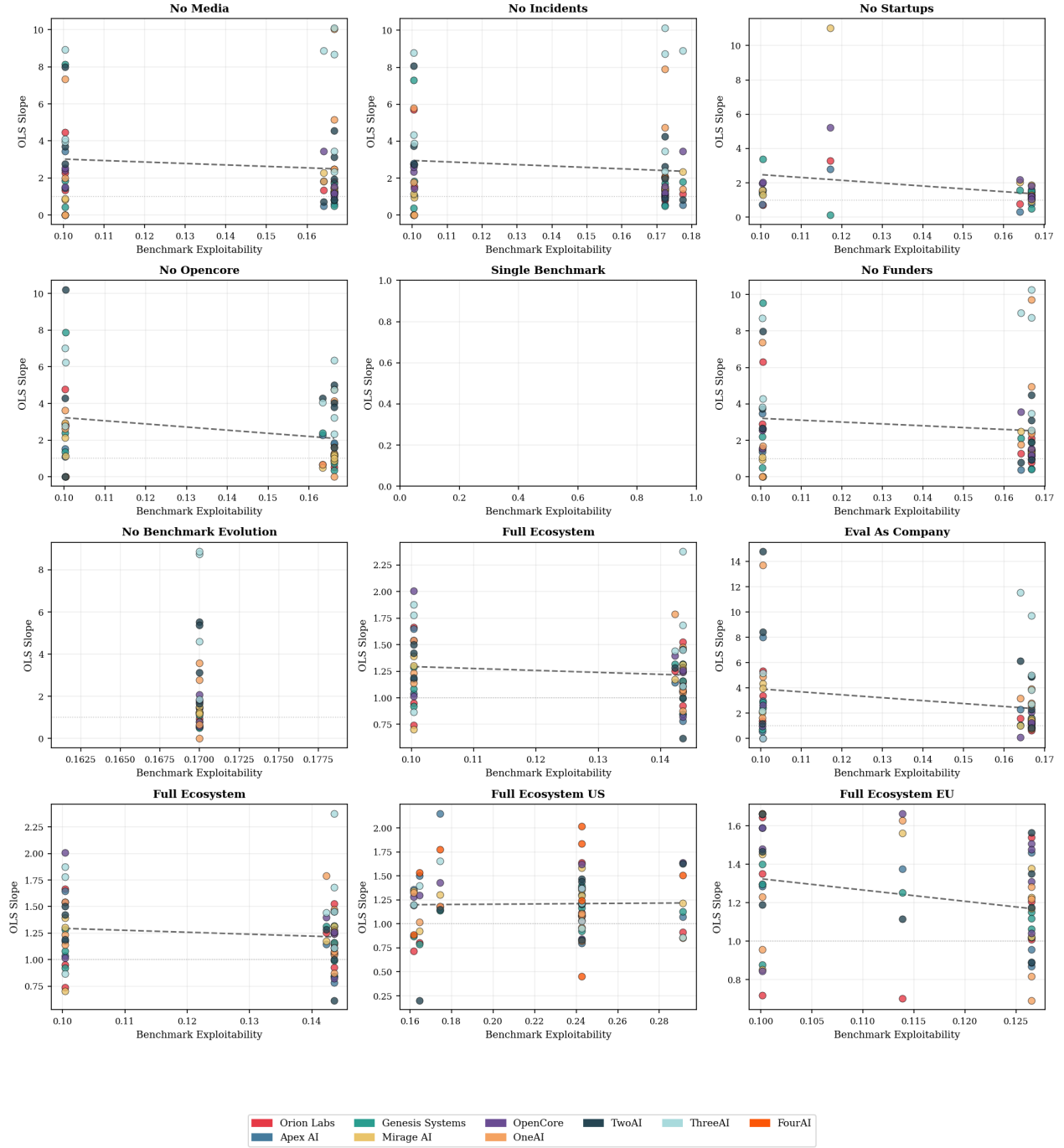


Figure 8: Regression slope vs. benchmark exploitability (β) across conditions. No positive association is observed, indicating that higher exploitability does not selectively amplify high-capability providers.

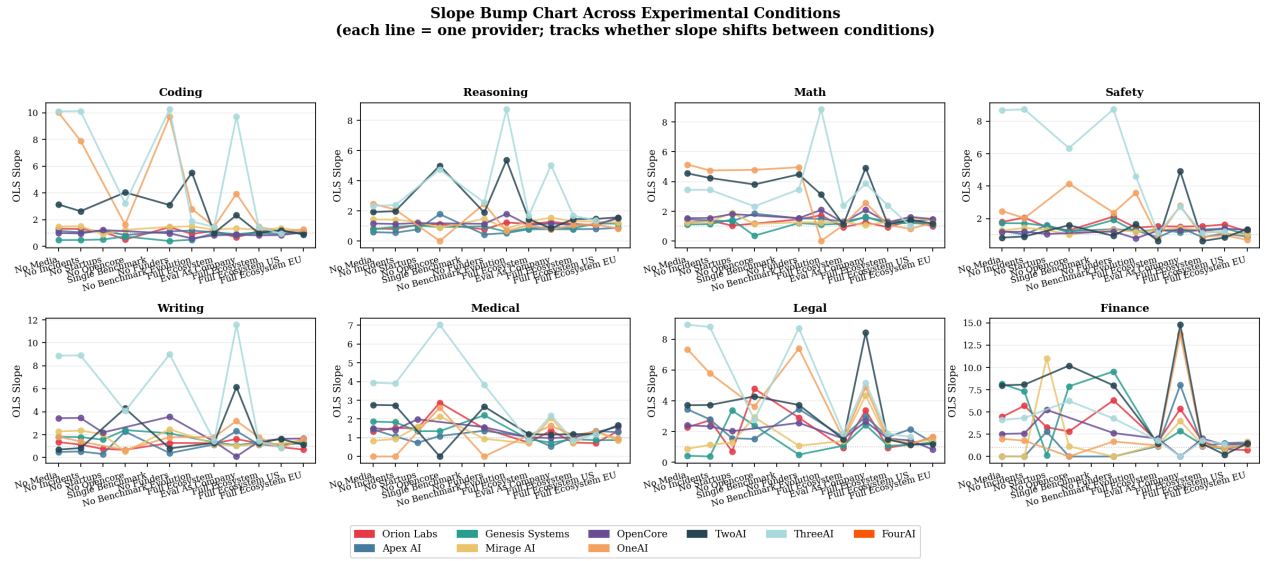


Figure 9: Condition-level variance in regression slope by domain. Finance and Legal exhibit the largest cross-condition variance, consistent with higher exploitability under full-ecosystem runs.